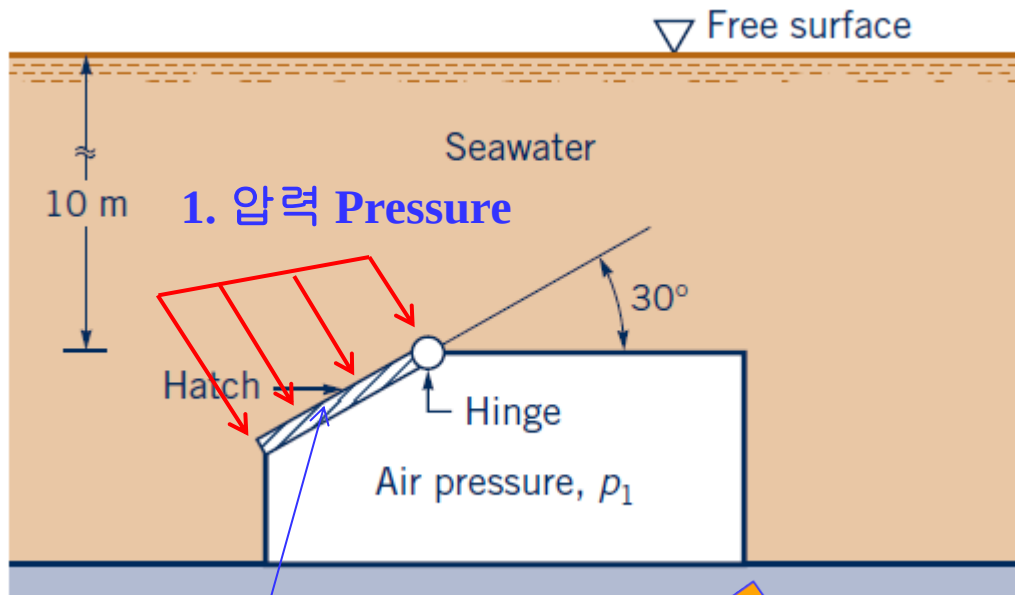


2장에서 공부할 주제들 Subjects to study in chapter 2



힘과 모멘트 Force & moment?

무게중심 centroid

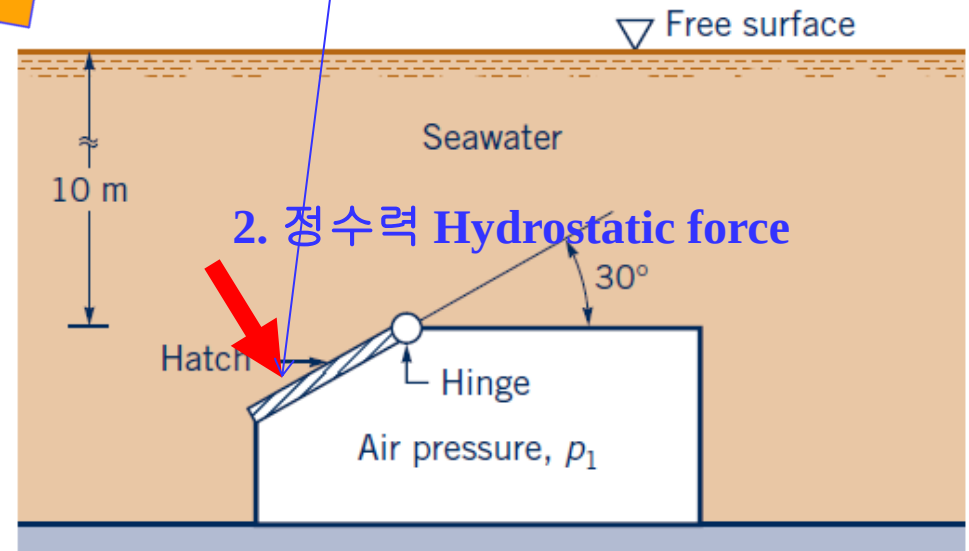
관성모멘트 moment of inertia

....

압력 Pressure ?

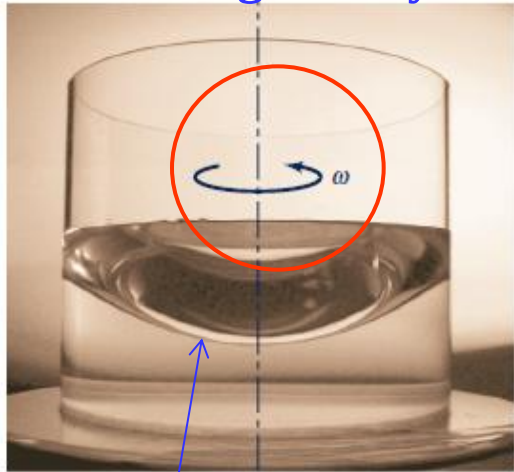
Pascal's law
measurement 측정

....

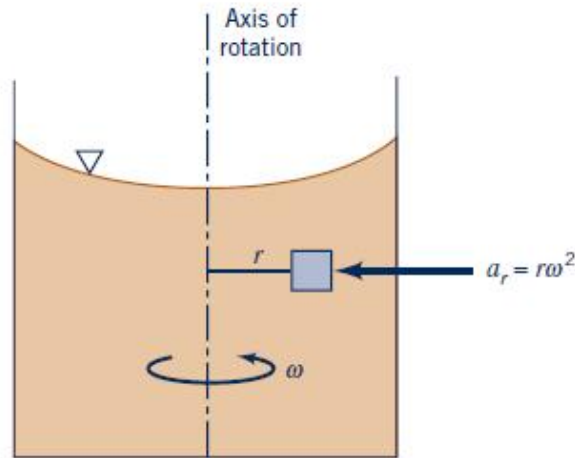


2장에서 공부할 주제들 Subjects to study in chapter 2

강체운동 Rigid body motion : 회전 rotation



(a)



(b)

■ **Figure 2.30** Rigid-body rotation of a liquid in a tank. (Photograph courtesy of Geno Pawlak.)

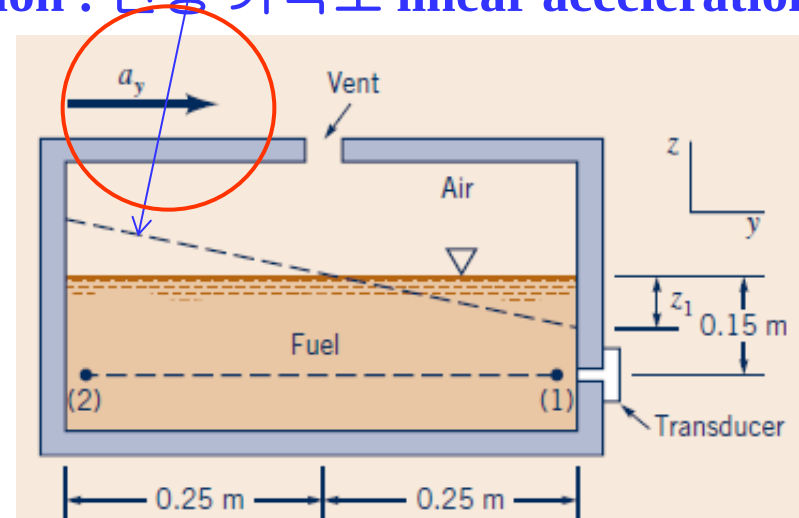
Free surface ?

hydrostatic pressure
body force
....

Rigid body motion : 선형 가속도 linear acceleration

자유표면 Free surface ?

정수압 hydrostatic pressure
체적력 body force
....



■ **Figure E2.11**

유체정역학 Fluid Statics

- contents

Pascal's law : pressure at a point

Pressure variation in static fluid

Pressure measurement & Manometry

Hydrostatic force on a plane surface

Pressure prism

Hydrostatic force on a curved surface

Buoyancy, Flotation and Stability

Rigid body motion

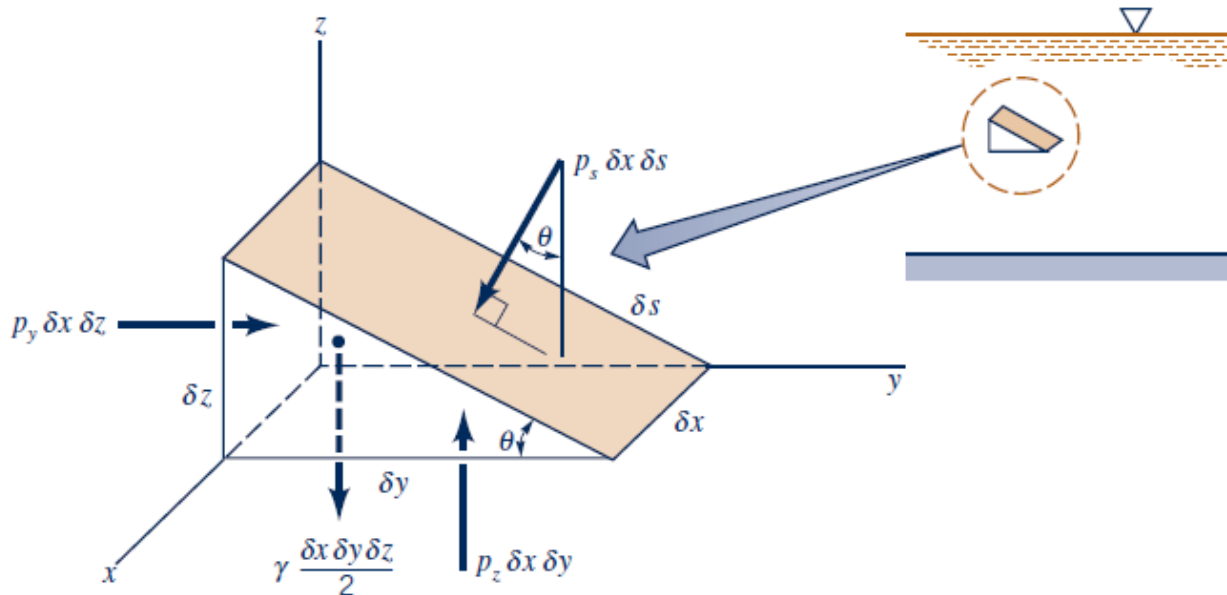


2.1 한 점에서의 압력 Pressure at a Point : 파스칼법칙 Pascal's law

- Pascal's law

정지 혹은 움직이는 유체 속 한 점에서의 압력은, 전단응력이 없는 한, 방향에 무관하다

The pressure at a point in a fluid at rest, or in motion, is independent of direction as long as there are no shearing stress present.



■ **Figure 2.1** Forces on an arbitrary wedge-shaped element of fluid.

2.1 한 점에서의 압력 Pressure at a Point : 파스칼법칙 Pascal's law

- 파스칼 법칙 Pascal's law

증명 proof) 정지하고 있는 작은 쐼기형 유체에 대하여

consider a small fluid wedge at rest

y-방향 힘의 평형 force equilibrium in the y-direction

$$\sum F_y = p_y(\delta x \cdot \delta z) - p_s(\delta x \cdot \delta s)\sin\theta = (p_y - p_s)(\delta x \cdot \delta z)$$

z-방향 힘의 평형 force equilibrium in the z-direction

$$\sum F_z = p_z(\delta x \cdot \delta y) - p_s(\delta x \cdot \delta s)\cos\theta = \frac{\rho}{2}(\delta x \cdot \delta y \cdot \delta z)g$$

since $\delta y = \delta s \cdot \cos\theta$ and $\delta z = \delta s \cdot \sin\theta$

$$\sum F_y = p_z(\delta x \cdot \delta z) - p_s(\delta x \cdot \delta s)\sin\theta = (p_z - p_s)(\delta x \cdot \delta z)$$

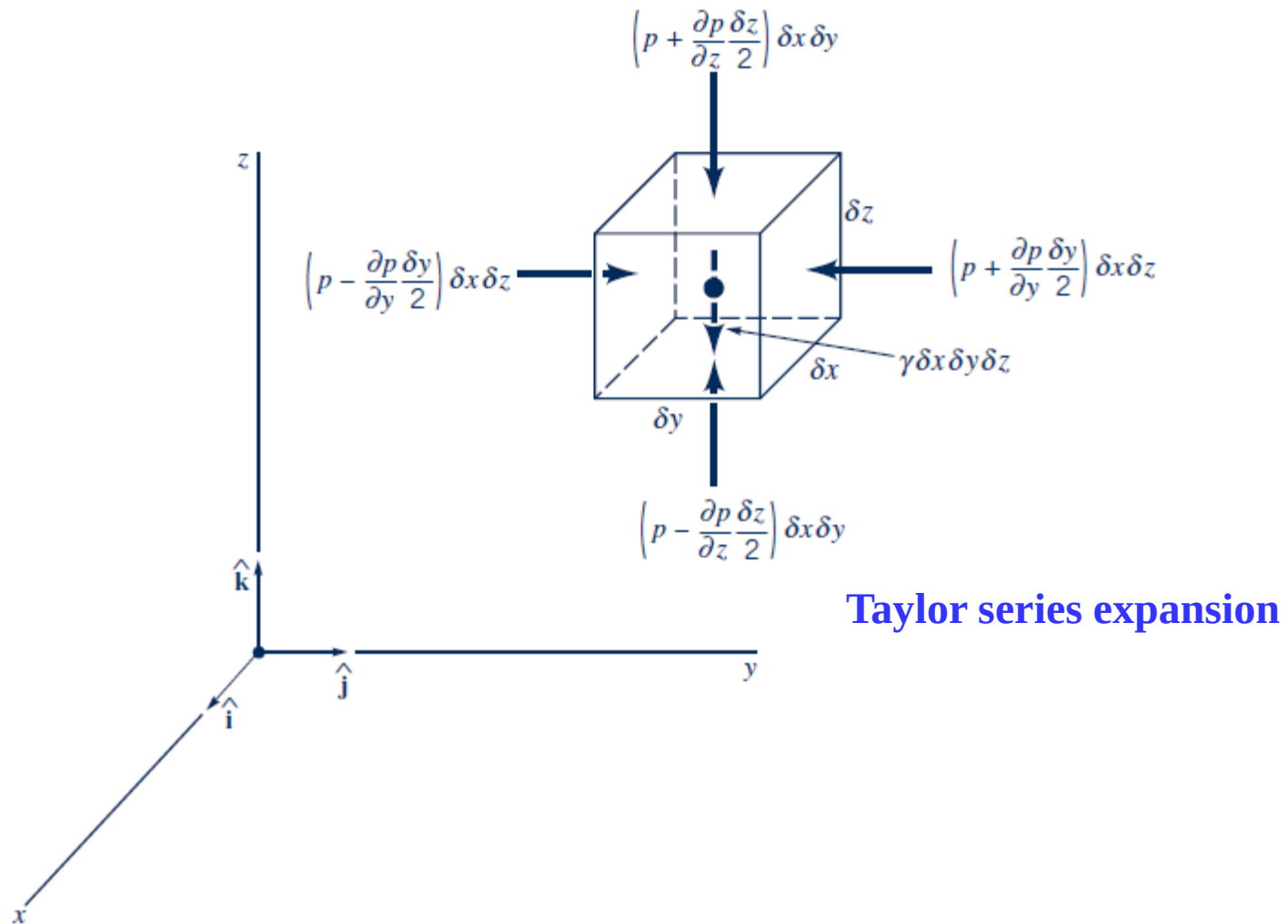
$$\therefore p_z - p_s = \frac{\rho}{2}(\delta y)g = 0 \text{ as } \delta x, \delta y, \delta z \rightarrow 0 \Rightarrow p_z = p_s$$

similarly, $p_z = p_s$

$$\therefore p_y = p_z = p_s$$

2.2 압력장에 대한 기본방정식 Basic Equation for Pressure field

- 뉴턴의 제2법칙 Newton's second law



■ **Figure 2.2** Surface and body forces acting on small fluid element.

2.2 압력장에 대한 기본방정식 Basic Equation for Pressure field

- 뉴턴의 제2법칙 **Newton's second law**

$$\sum \delta \vec{F} = \delta m \cdot \vec{a}$$

- y방향으로 **In the y direction**

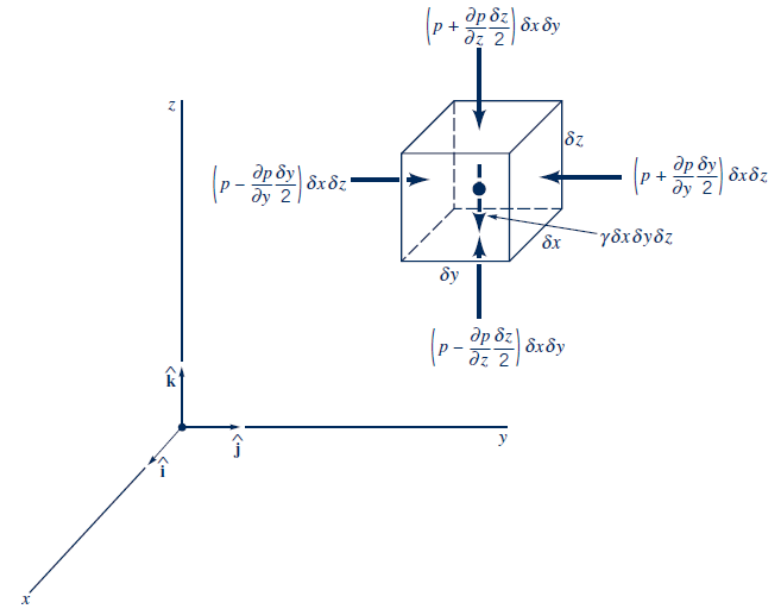
$$\sum \delta F_y = \delta m \cdot a_y = \rho(\delta x \cdot \delta y \cdot \delta z) a_y$$

$$\sum \delta F_y = \sum \delta F_{surface,y} + \sum \delta F_{body,y}$$

$$\sum \delta F_{body,y} = \text{gravitational force} = \rho \cdot g_y \cdot \delta x \cdot \delta y \cdot \delta z$$

$$\begin{aligned} \sum \delta F_{surface,y} &= \left(p - \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \cdot \delta z - \left(p + \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \cdot \delta z \\ &= -\frac{\partial p}{\partial y} \delta x \cdot \delta y \cdot \delta z \end{aligned}$$

$$\therefore -\frac{\partial p}{\partial y} \delta x \cdot \delta y \cdot \delta z + \rho \cdot g_y \cdot \delta x \cdot \delta y \cdot \delta z = \rho(\delta x \cdot \delta y \cdot \delta z) a_y$$



■ **Figure 2.2** Surface and body forces acting on small fluid element.

$$\therefore -\frac{\partial p}{\partial y} + \rho \cdot g_y = \rho a_y$$

2.2 압력장에 대한 기본방정식 Basic Equation for Pressure field

- x방향으로 In the x direction

$$\sum \delta F_x = \delta m \cdot a_x = \rho(\delta x \cdot \delta y \cdot \delta z)a_x$$

$$\therefore -\frac{\partial p}{\partial x} + \rho \cdot g_x = \rho a_x$$

- z방향으로 In the z direction

$$\sum \delta F_z = \delta m \cdot a_z = \rho(\delta x \cdot \delta y \cdot \delta z)a_z$$

$$\therefore -\frac{\partial p}{\partial z} + \rho \cdot g_z = \rho a_z$$

- 벡터로 표현하면 In a vector form

$$\sum \delta \vec{F} = \delta m \cdot \vec{a} = \rho(\delta x \cdot \delta y \cdot \delta z)\vec{a}$$

$$\therefore -\nabla p + \rho \cdot \vec{g} = \rho \cdot \vec{a}$$

$$\nabla p = \frac{\partial p}{\partial x} \vec{i} + \frac{\partial p}{\partial y} \vec{j} + \frac{\partial p}{\partial z} \vec{k} \quad \vec{g} = g_x \vec{i} + g_y \vec{j} + g_z \vec{k} \quad \vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

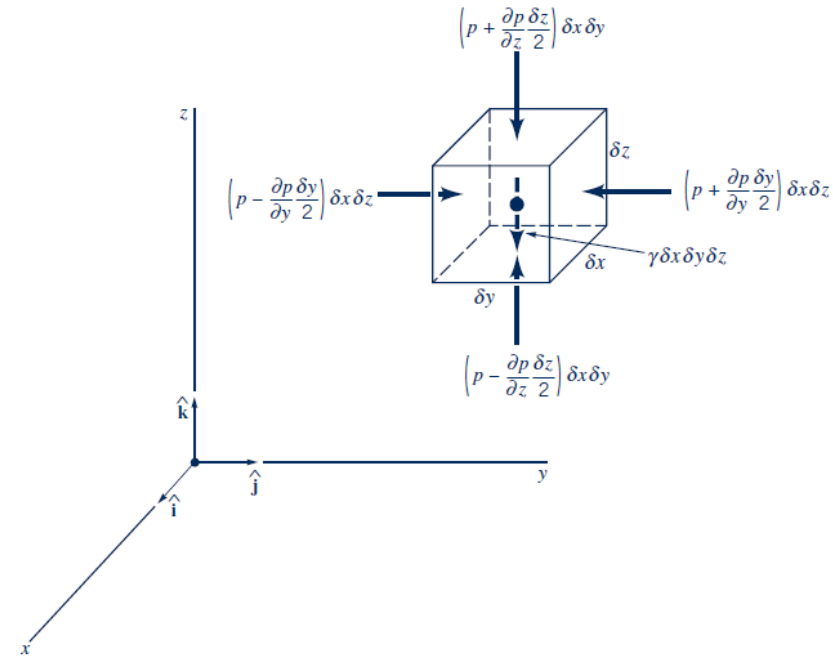


Figure 2.2 Surface and body forces acting on small fluid element.

2.3 정지된 유체 내 압력 변화 Pressure Variation in a Fluid at Rest

- 정지상태에서 At rest

$$\vec{a} = 0$$

$$\therefore -\nabla p + \rho \cdot \vec{g} = 0$$

- ✓ 중력이 (-)z 방향이라면 If the gravitational force is in the negative z direction

$$\vec{g} = -g\vec{k}$$

$$-\nabla p - \rho g\vec{k} = 0$$

$$\therefore \nabla p = \frac{\partial p}{\partial x}\vec{i} + \frac{\partial p}{\partial y}\vec{j} + \frac{\partial p}{\partial z}\vec{k} = -\rho g\vec{k}$$

$$\therefore \frac{\partial p}{\partial x} = \frac{\partial p}{\partial y} = 0 \text{ and } \frac{\partial p}{\partial z} = -\rho g = -\gamma$$

압력 P는 z 방향으로만 바뀐다(변한다) P depends only on z

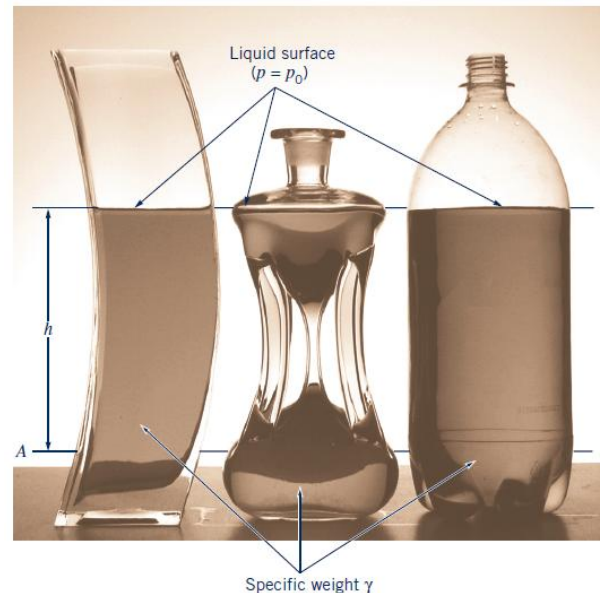
2.3 정지된 유체 내 압력 변화 Pressure Variation in a Fluid at Rest

- 2.3.1 비압축성 유체 Incompressible Fluid

$$\frac{\partial p}{\partial z} = -\rho g = -\gamma \quad \int_{p_1}^{p_2} dp = - \int_{z_1}^{z_2} \rho g dz$$
$$\int_{p_1}^{p_2} dp = p_2 - p_1$$
$$-\rho g \int_{z_1}^{z_2} dz = -\rho g(z_2 - z_1) = -\gamma(z_2 - z_1)$$
$$\therefore p_2 - p_1 = -\gamma(z_2 - z_1)$$

- 임의 형상 용기 속 압력 Pressure in containers of arbitrary shape

$$p = p_0 + \gamma h$$



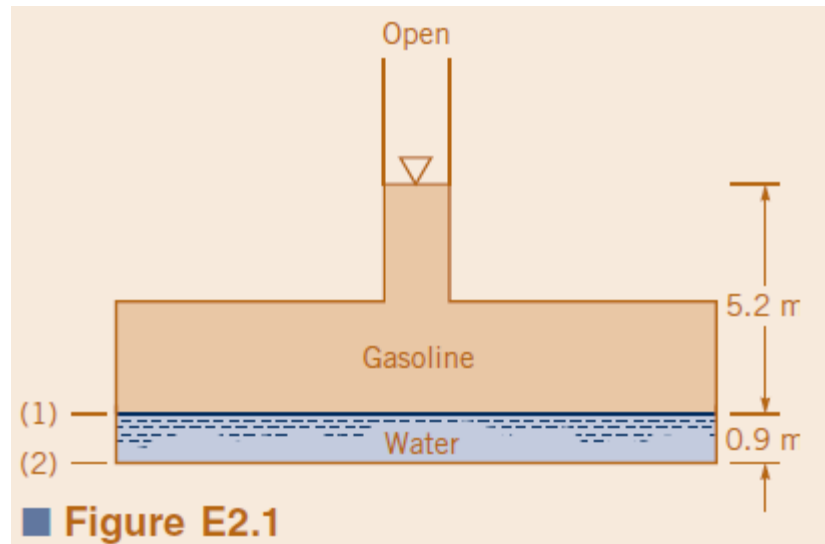
■ Figure 2.4 Fluid pressure in containers of arbitrary shape.

2.3 정지된 유체 내 압력 변화 Pressure Variation in a Fluid at Rest

- Ex 2.1) Determine the pressure at the gasoline-water interface and at the bottom of the tank. Express the pressure in units of $\text{Pa}(\text{N/m}^2)$, N/mm^2 , and as a pressure head in meters of water. The specific gravity of gasoline is $\text{SG}=0.68$.

가솔린과 물의 경계면과 바닥에서의 압력을 $\text{Pa}(\text{N/m}^2)$, N/mm^2 그리고 물 (높이)를 미터로 (표시)하여 압력수두의 단위로 구하라. 가솔린의 비중은 $\text{SG}=0.68$ 이다.

$$p_1, \quad p_2?$$



2.3 정지된 유체 내 압력 변화 Pressure Variation in a Fluid at Rest

- 가솔린과 물의 경계면에서의 압력 Pressure at the gasoline-water interface

$$p_1 = p_0 + \gamma_{gasoline}h = p_0 + 0.68 \times 1000 \left(\frac{kN}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right) 5.2(m)$$

$$= p_0 + 34.7 \left(\frac{kN}{m^2} \right)$$

$$\therefore p_{1,gage} = 34.7 \left(\frac{kN}{m^2} \right) = \frac{34.7 \left(\frac{kN}{m^2} \right)}{10^6 \left(\frac{m^2}{mm^2} \right)} = 0.044 \left(\frac{N}{mm^2} \right)$$

$$\therefore head = \frac{p_{1,gage}}{\gamma_{water}} = \frac{34.7 \left(\frac{kN}{m^2} \right)}{1000 \left(\frac{kg}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right)} = 3.54(m)$$

- 탱크 바닥에서의 압력 Pressure at the bottom of the tank

$$\therefore p_{2,gage} = p_{1,gage} + \gamma_{water}h = 34.7 \left(\frac{kN}{m^2} \right) + 1000 \left(\frac{kg}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right) 0.9(m)$$

$$= 43.5 \left(\frac{kN}{m^2} \right) = \frac{43.5 \left(\frac{kN}{m^2} \right)}{10^6 \left(\frac{m^2}{mm^2} \right)} = 0.044 \left(\frac{N}{mm^2} \right)$$

$$\therefore head = \frac{p_{2,gage}}{\gamma_{water}} = \frac{43.5 \left(\frac{kN}{m^2} \right)}{1000 \left(\frac{kg}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right)} = 4.44(m)$$

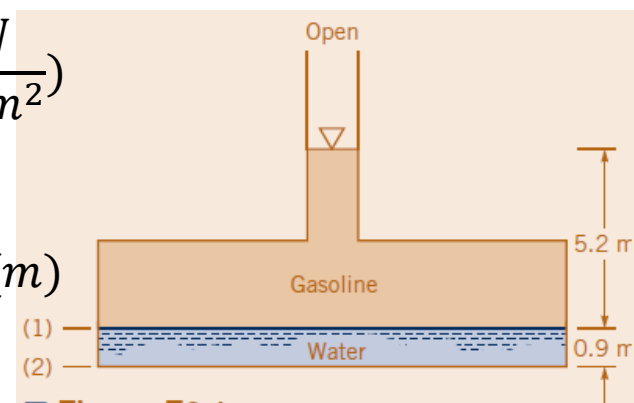


Figure E2.1

2.3.2 압축성 유체 Compressible Fluid

- 이상기체에 대하여 For an ideal gas

$$\rho = \frac{P}{RT} \text{ and } \frac{dp}{dz} = -\rho g = -\frac{pg}{RT}$$

$$\therefore \int_{p_1}^{p_2} \frac{dp}{p} = - \int_{z_1}^{z_2} \frac{g}{RT} dz \Rightarrow \ln \frac{p_2}{p_1} = -\frac{g}{R} \int_{z_1}^{z_2} \frac{g}{T} dz$$

등온조건이라면 if isothermal condition, i.e. $T=\text{constant}$

$$\ln \frac{p_2}{p_1} = -\frac{g}{R} \int_{z_1}^{z_2} \frac{g}{T} dz = -\frac{g}{RT} (z_2 - z_1)$$

$$\therefore p_2 = p_1 e^{-\frac{g(z_2 - z_1)}{RT_0}}$$

2.4 표준대기 Standard Atmosphere

- 표준대기는 지구 대기에서의 평균 조건을 이상적으로 표현한 것이다

The standard atmosphere is an idealized representation of mean conditions in the earth's atmosphere

$$T = T_a - \beta z$$

in the troposphere

$$\beta = \text{lapse rate} = 0.0065 \left(\frac{K}{m} \right)$$

$$\ln \frac{p_2}{p_1} = -\frac{g}{R} \int_{z_1}^{z_2} \frac{g}{T} dz = -\frac{g}{R} \int_{z_1}^{z_2} \frac{g}{T_a - \beta z} dz$$

$$\therefore p = p_a \left(1 - \frac{\beta z}{T_a} \right)^{g/R\beta}$$

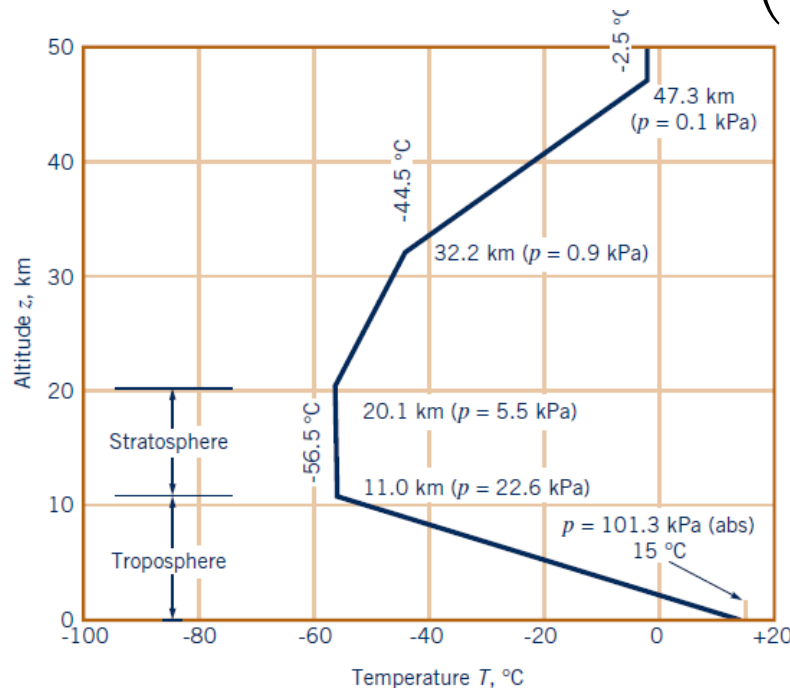
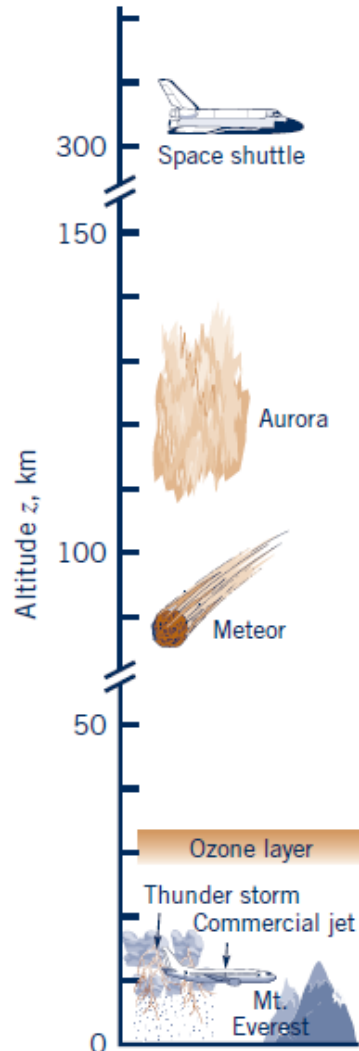


Figure 2.6 Variation of temperature with altitude in the U.S. standard atmosphere.

2.5 압력측정 Measurement of Pressure

- 절대, 게이지, 그리고 진공압력 Absolute, gage, and vacuum pressure

게이지 압력 = 절대압력 - 대기압

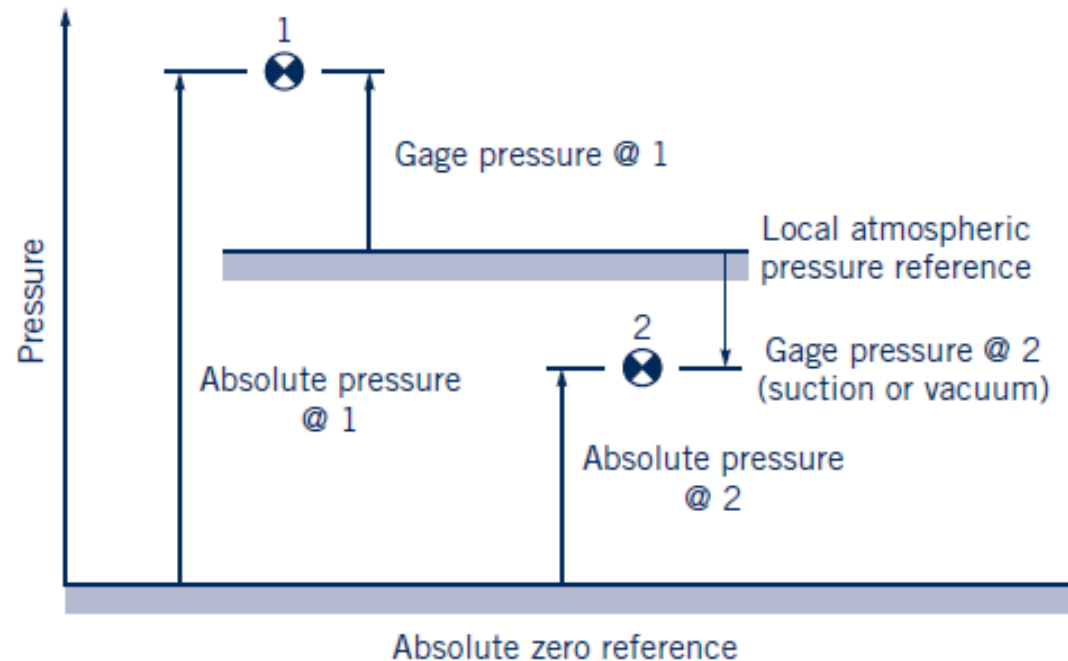
gage pressure = absolute pressure – atmospheric pressure

진공압력 = 대기압 – 절대압력

vacuum pressure = atmospheric pressure – absolute pressure

$$p_{gage} = p_{absolute} - p_{atmospheric}$$

$$p_{vacuum} = p_{atmospheric} - p_{absolute}$$

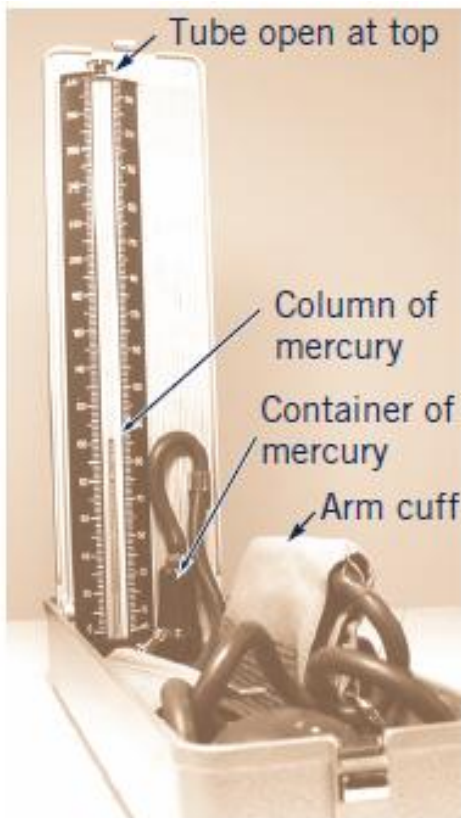


■ **Figure 2.7** Graphical representation of gage and absolute pressure.

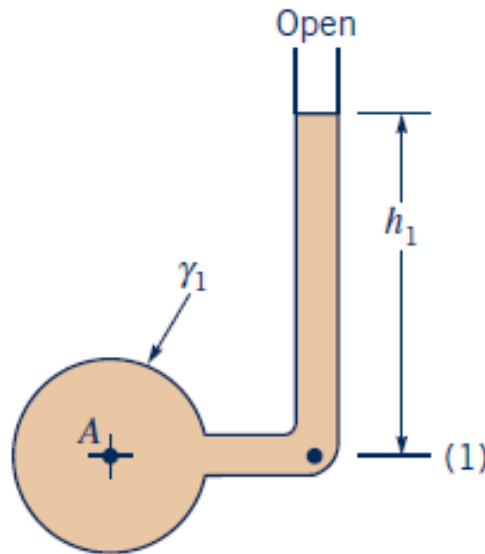
2.6 액주계 Manometry

- 액주계는 압력을 측정하기 위해 수직 혹은 경사진 액체기둥을 이용한다
Manometers use vertical or inclined liquid columns to measure pressure
- 2.6.1 Piezometer Tube 피에조 관
 - sphygmomanometer
 - piezometer tube

$$p = p_0 + \gamma h$$



$$p_{A,gage} = \gamma h$$



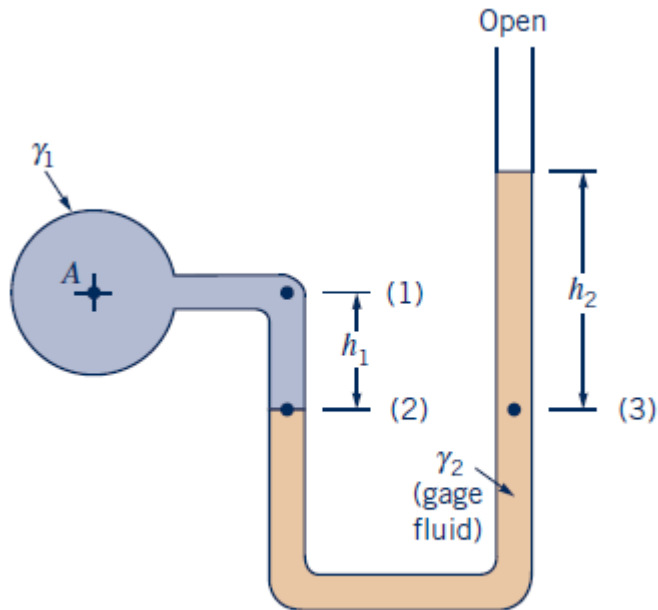
■ Figure 2.9 Piezometer tube.

2.6.2 U-자형 액주계 U-Tube Manometer

- 단순 U자형 마노미터 Simple U-Tube Manometers

$$p_{A,gage} + \gamma_1 h_1 - \gamma_2 h_2 = 0$$

$$\therefore p_{A,gage} = \gamma_2 h_2 - \gamma_1 h_1$$



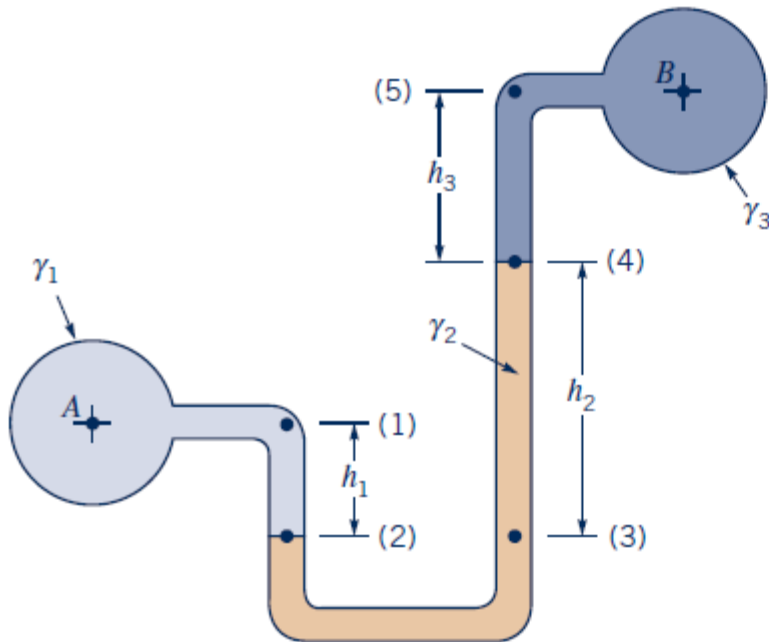
■ **Figure 2.10** Simple U-tube manometer.

2.6.2 U-튜브 액주계 U-Tube Manometer

- U-튜브 시차 액주계 Differential U-Tube Manometers

$$p_{A,gage} + \gamma_1 h_1 - \gamma_2 h_2 - \gamma_3 h_3 = p_{B,gage}$$

$$\therefore p_{A,gage} - p_{B,gage} = \gamma_2 h_2 + \gamma_3 h_3 - \gamma_1 h_1$$



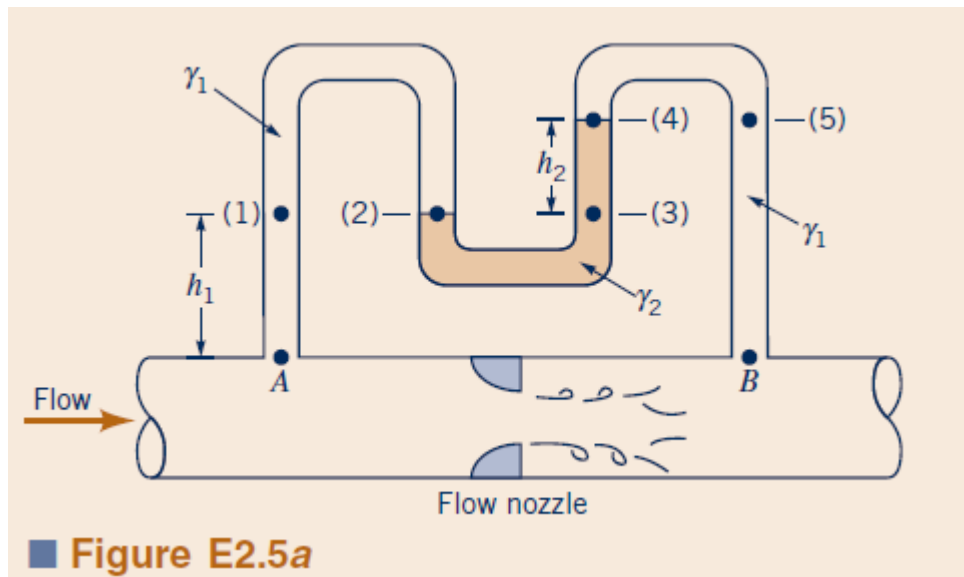
■ Figure 2.11 Differential U-tube manometer.

Example : 2.6.2 U-Tube Manometer

- Ex 2.5) (a) Determine an equation for $p_A - p_B$ in terms of the specific weight of the flowing fluid, γ_1 , the specific weight of the gauge fluid, γ_2 and the various heights indicated. (b) For $\gamma_1 = 9.80(\text{kN/m}^3)$, $\gamma_2 = 15.6(\text{kN/m}^3)$, $h_1 = 1.0(\text{m})$, and $h_2 = 0.5(\text{m})$, what is the value of the pressure drop, $p_A - p_B$?

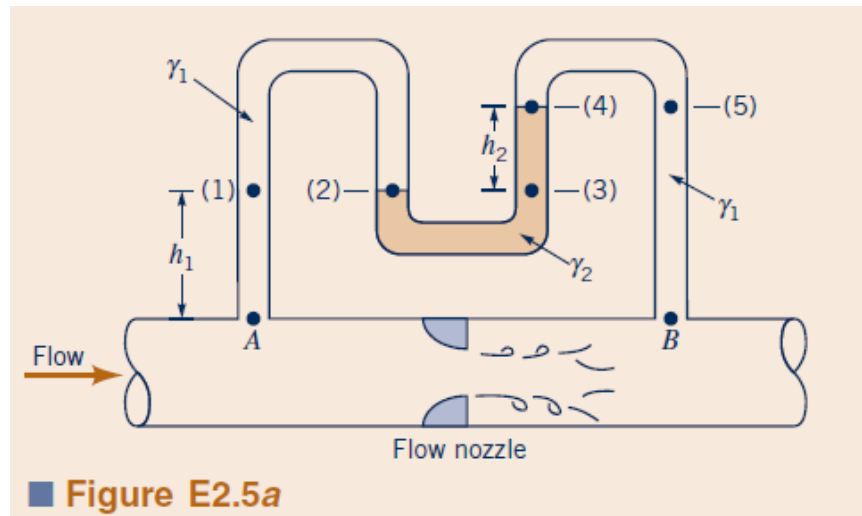
(a) $p_A - p_B$ 를 유동하는 유체의 비중량 γ_1 , 계기 유체의 비중량 γ_2 , 그리고 다양한 높이의 함수로 구하라.

(b) $\gamma_1 = 9.80(\text{kN/m}^3)$, $\gamma_2 = 15.6(\text{kN/m}^3)$, $h_1 = 1.0(\text{m})$, and $h_2 = 0.5(\text{m})$ 일 경우 $p_A - p_B$ 를 구하라



Example : 2.6.2 U-Tube Manometer

- Ex 2.5) (a) Determine an equation for $p_A - p_B$ in terms of the specific weight of the flowing fluid, γ_1 , the specific weight of the gauge fluid, γ_2 and the various heights indicated. (b) For $\gamma_1 = 9.80(\text{kN/m}^3)$, $\gamma_2 = 15.6(\text{kN/m}^3)$, $h_1 = 1.0(\text{m})$, and $h_2 = 0.5(\text{m})$, what is the value of the pressure drop, $p_A - p_B$?



$$a) p_{A,gage} - \gamma_1 h_1 - \gamma_2 h_2 + \gamma_1 (h_1 + h_2) = p_{B,gage}$$

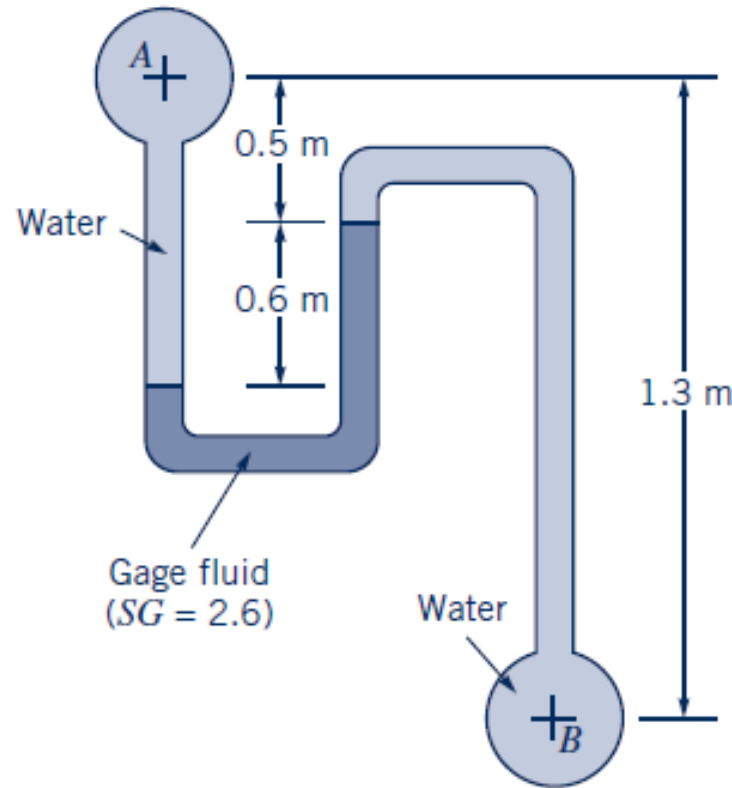
$$\therefore p_{A,gage} - p_{B,gage} = h_2 (\gamma_2 - \gamma_1)$$

$$b) p_{A,gage} - p_{B,gage} = h_2 (\gamma_2 - \gamma_1) = 0.5(\text{m})(15.6 - 9.8) \left(\frac{\text{kN}}{\text{m}^3} \right) = 2.9(\text{kPa})$$



Quiz #1: U-Tube Manometer

- Q1) Two pipes are connected by a manometer as shown below. Determine the pressure difference, $P_A - P_B$, between the pipes. 두 개의 관이 그림과 같이 액주계로 연결되어 있다. 두 관 사이의 압력차 $P_A - P_B$ 를 구하라. (10점)



■ Figure P2.40



Quiz #2: U-Tube Manometer

Q2) The sensitivity Sen of the micromanometer shown below is defined as

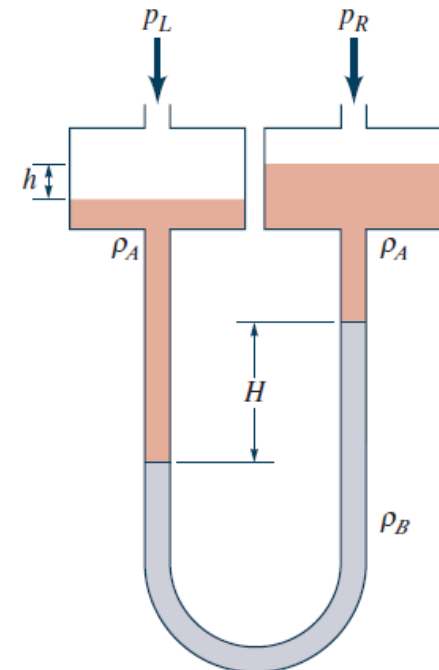
$$Sen = \frac{H}{p_L - p_R}$$

Find the sensitivity of the micromanometer in terms of the densities. How can the sensitivity be increased?.

그림과 같은 마이크로 액주계의 민감도 (sen)는 다음과 같이 정의된다.

$$Sen = \frac{H}{p_L - p_R}$$

마이크로 액주계의 민감도를 밀도로 표현하라.
어떻게 이 민감도를 높일 수 있는가? (10점)

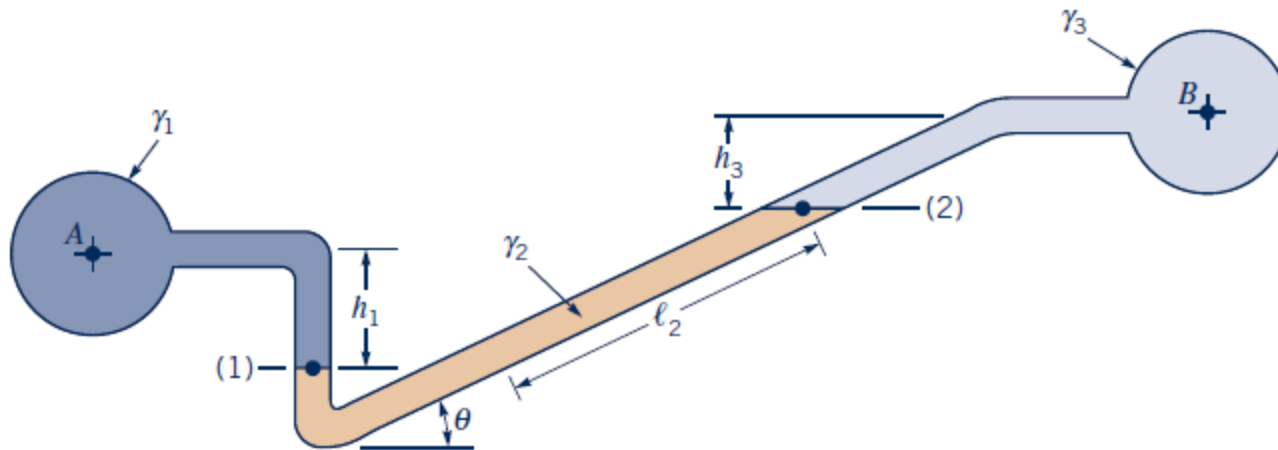


2.6.3 경사관 액주체 Inclined Tube Manometer

- Inclined Tube Manometers : useful to measure small differences in gas pressure. 작은 기체 압력 차이를 측정하는데 유용

$$p_{A,gage} + \gamma_1 h_1 - \gamma_2 l_2 \sin\theta - \gamma_3 h_3 = p_{B,gage}$$

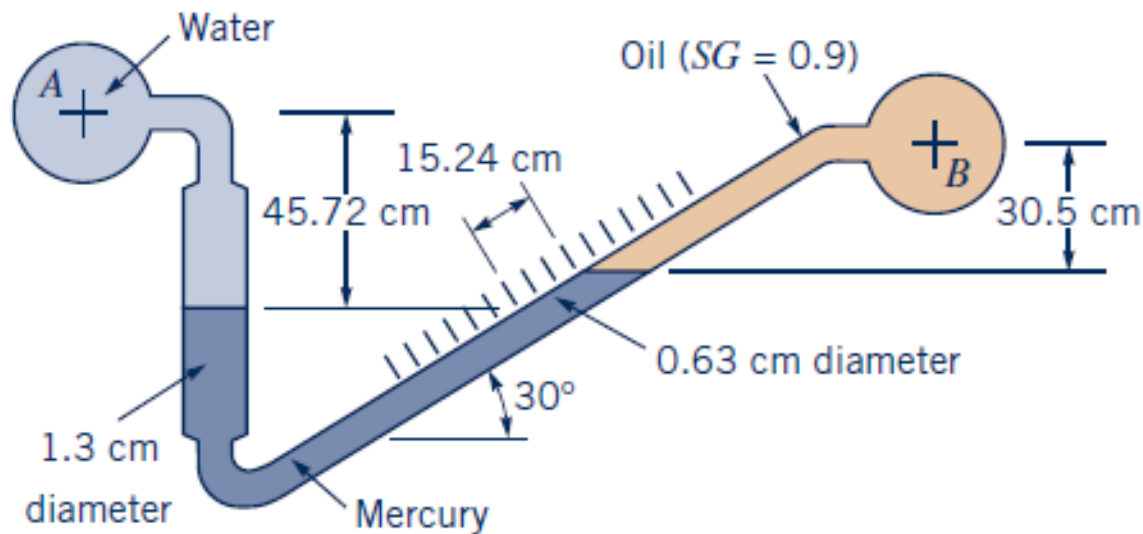
$$\begin{aligned} \therefore p_{A,gage} - p_{B,gage} &= \gamma_2 l_2 \sin\theta + \gamma_3 h_3 - \gamma_1 h_1 \\ &\approx \gamma_2 l_2 \sin\theta \text{ if } \gamma_2 \gg \gamma_1, \gamma_3 \end{aligned}$$



■ **Figure 2.12** Inclined-tube manometer.

Quiz #3: Review Problem

- Q3) Determine the change in the elevation of the mercury in the left leg of the manometer of the figure as a result of an increase in pressure of 34.4(kPa) in pipe A while the pressure in pipe B remains constant. 파이프 B의 압력이 일정할 때 파이프 A의 압력이 34.4(kPa) 증가하는 경우 액주계의 왼쪽 수은 눈금이 얼마나 올라가는지 구하라.

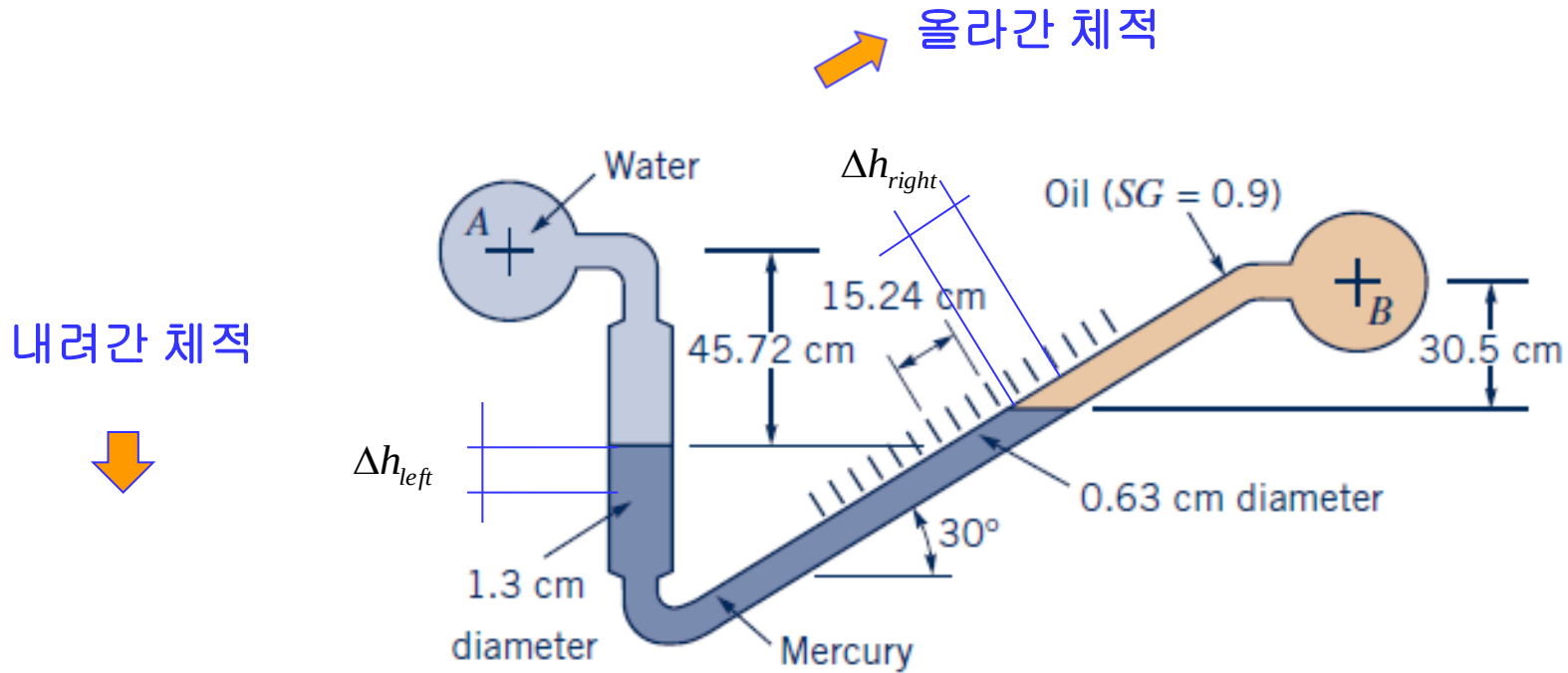


$$p_A = 34.4(\text{kPa})$$

■ Figure P2.64

Quiz #3: Review Problem

- 힌트



■ Figure P2.64

2.7 기계식 및 전자식 압력 측정 장치

Mechanical and Electronic Pressure Measuring Devices

- 디지털 마노미터 Digital manometer



- Classic manometer

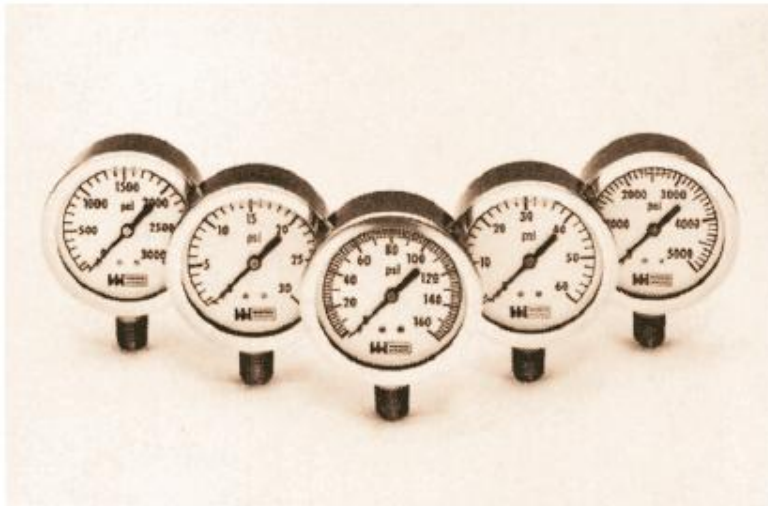
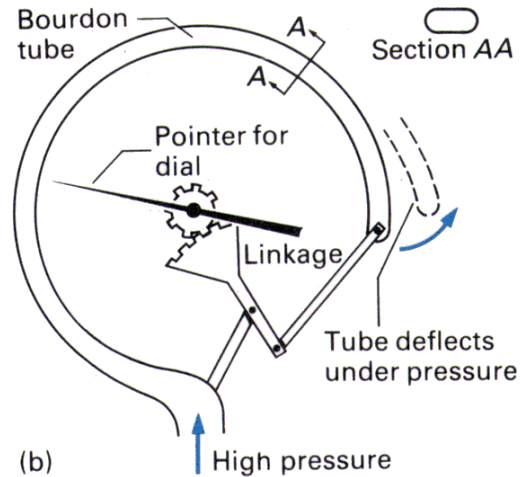
Fi

전통 액주계를 여전히 사용하는 분야?
어떤 이유로?

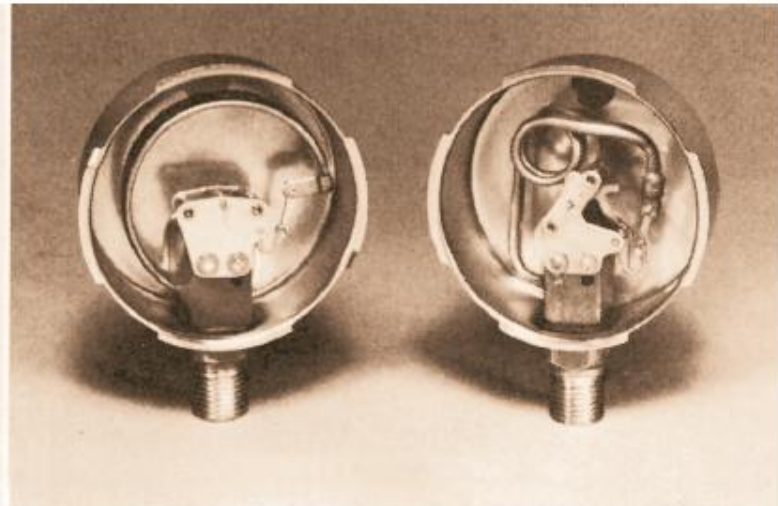
2.7 기계식 및 전자식 압력 측정 장치

Mechanical and Electronic Pressure Measuring Devices

- Bourdon gage



(a)



(b)

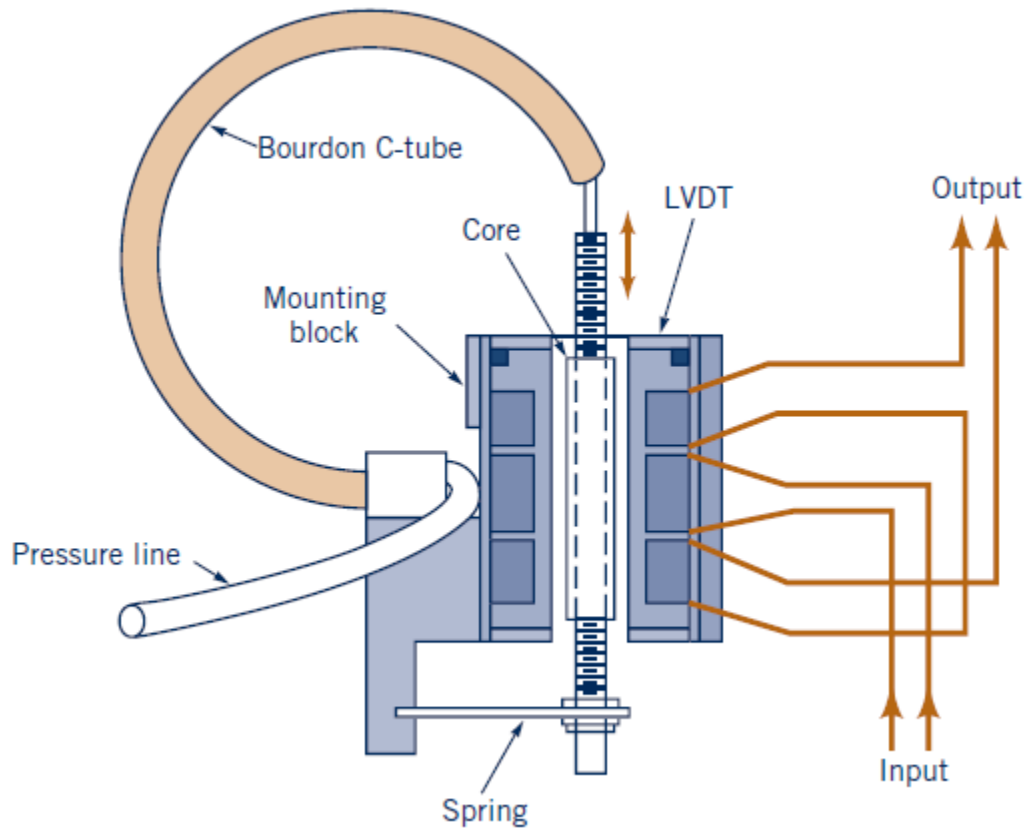
Figure 2.13 (a) Liquid-filled Bourdon pressure gages for various pressure ranges. (b) Internal elements of Bourdon gages. The “C-shaped” Bourdon tube is shown on the left, and the “coiled spring” Bourdon tube for high pressures of 6895 kPa and above is shown on the right.

(Photographs courtesy of Weiss Instruments, Inc.)

2.7 기계식 및 전자식 압력 측정 장치

Mechanical and Electronic Pressure Measuring Devices

- Pressure transducer

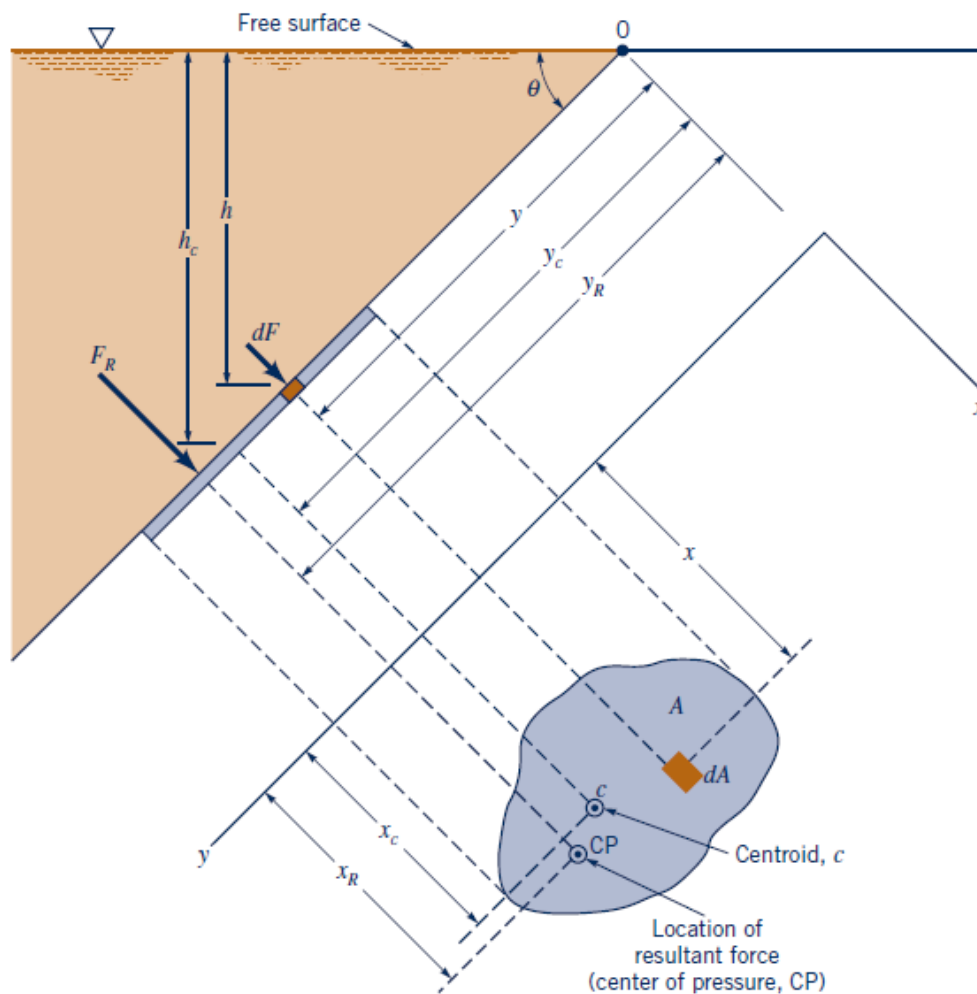


■ **Figure 2.14** Pressure transducer that combines a linear variable differential transformer (LVDT) with a Bourdon gage. (From Ref. 4, used by permission.)

2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- 유체 속에 잠긴 평면에 작용하는 합력

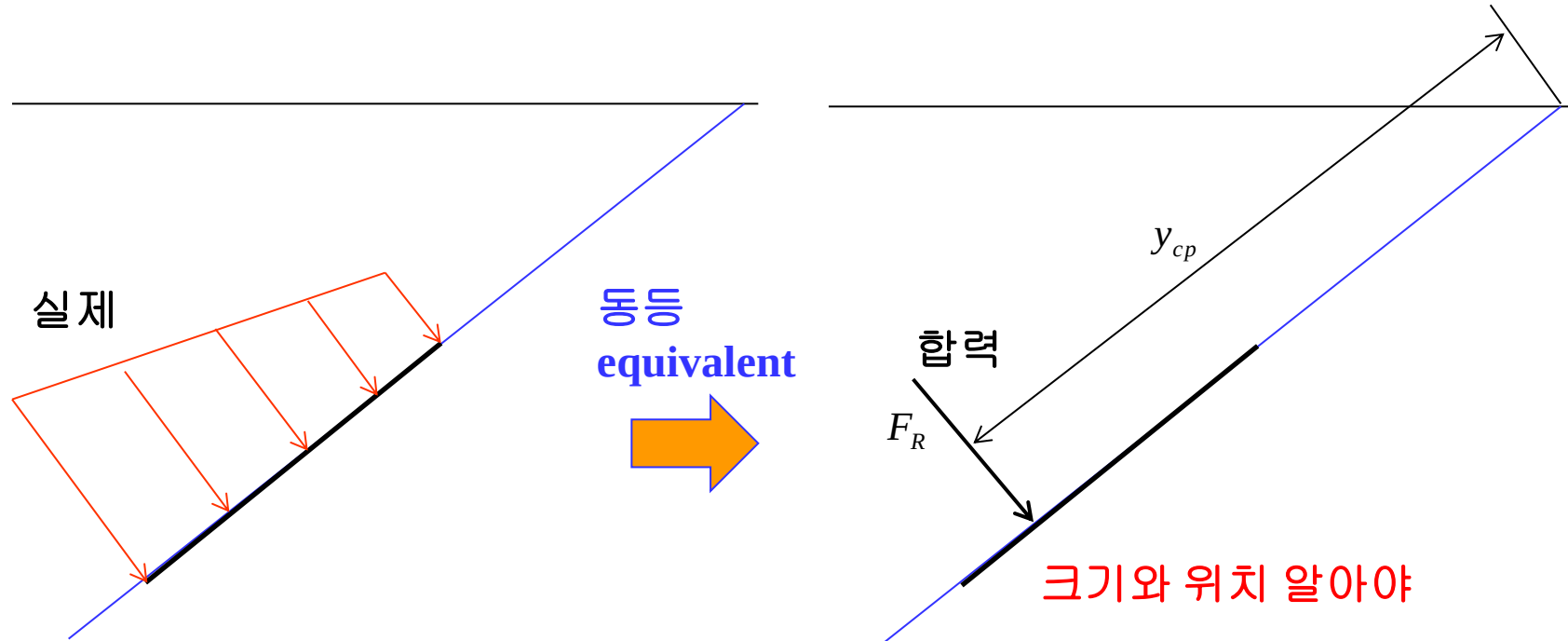
Resultant force on a plane surface submerged in a fluid



■ **Figure 2.17** Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- Resultant force on a plane surface submerged in a fluid



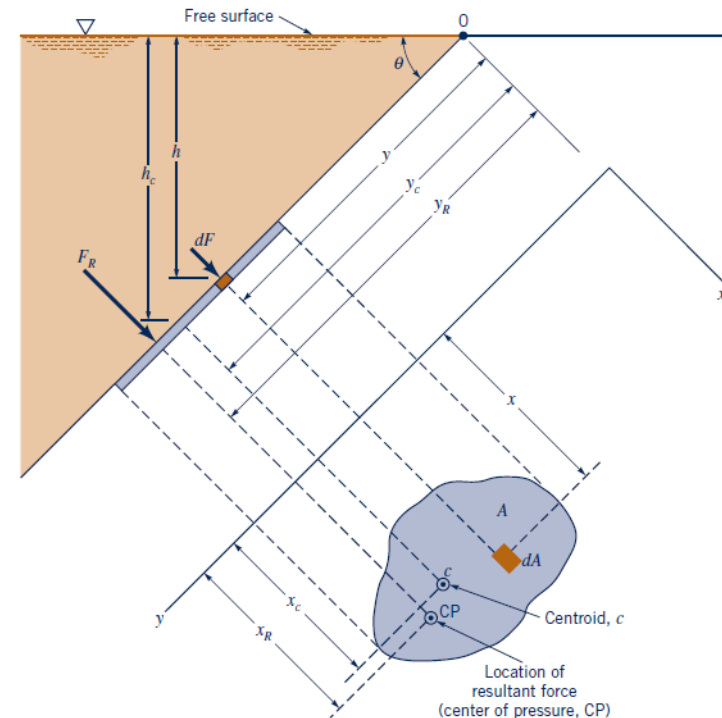
2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- Resultant force on a plane surface submerged in a fluid

$$\begin{aligned}
 F_R &= \int dF = \int p dA = \int \gamma \cdot h dA = \int \gamma \cdot y(\sin\theta) dA = \gamma \cdot \sin\theta \int y dA \\
 &= (\gamma \cdot \sin\theta) y_c \cdot A = \gamma \cdot h_c \cdot A = \mathbf{p_c \cdot A}
 \end{aligned}$$

- 단면 1차 모멘트 First moment of the area = centroid

$$y_c = \frac{\int y dA}{A}$$



■ **Figure 2.17** Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- 합력의 위치 Location of resultant force : 압력중심 center of pressure

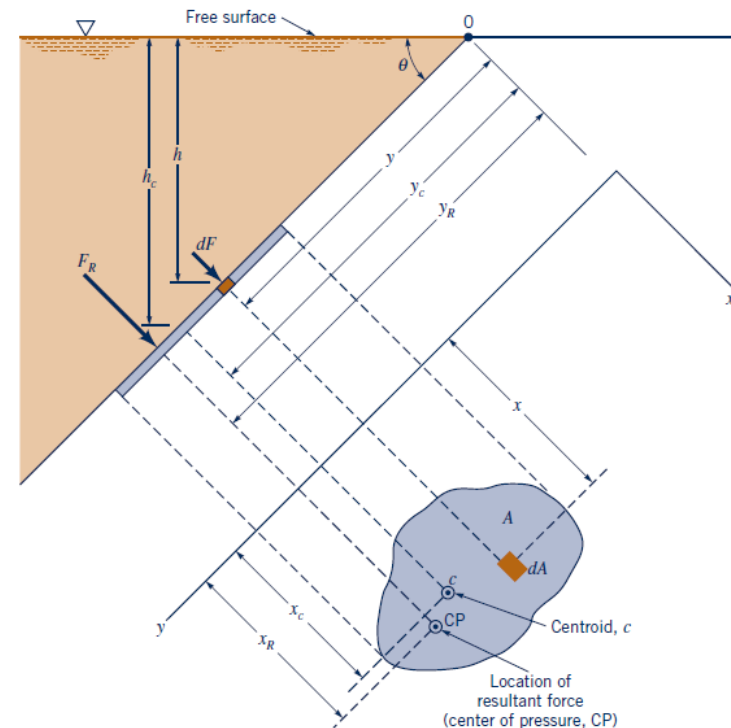
$$F_R y_R = \int y dF = \int y(\gamma \cdot y \cdot \sin\theta) dA = \int (\gamma \cdot y^2 \cdot \sin\theta) dA$$

$$= \gamma \cdot \sin\theta \int (y^2) dA$$

$$\begin{aligned} \therefore y_R &= \frac{\gamma \cdot \sin\theta \int (y^2) dA}{F_R} = \frac{\gamma \cdot \sin\theta \int (y^2) dA}{\gamma \cdot \sin\theta \cdot y_c \cdot A} \\ &= \frac{\int (y^2) dA}{y_c \cdot A} = \frac{I_x}{y_c \cdot A} \end{aligned}$$

- 단면 2차 모멘트 Second moment of the area

$$I_x = I_{yy} = \int (y^2) dA$$



■ **Figure 2.17** Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- 평행축 정리 Parallel axis theorem

$$I_x = \int (y^2) dA = \int (y_c + y')^2 dA = \int (y_c)^2 + 2y_c \cdot y' + (y')^2 dA$$

$$= Ay_c^2 + \int (y')^2 dA = Ay_c^2 + I_{xc}$$

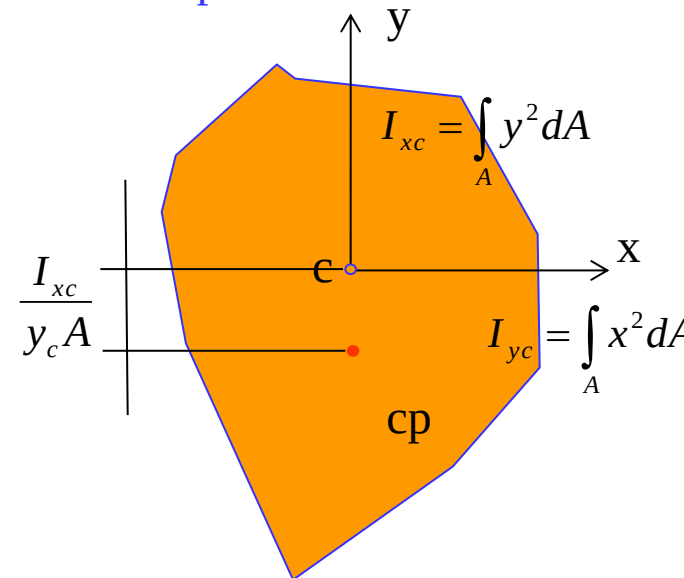
I_{xc} is the second moment of the area with respect to an axis passing through its centroid 좌표의 중심을 무게중심에 두었을 때의 단면 2차 모멘트

- 합력의 위치 Location of resultant force : 압력중심 center of pressure

$$y_R = \frac{\int (y^2) dA}{y_c \cdot A} = \frac{I_x}{y_c \cdot A} = \frac{I_{xc} + Ay_c^2}{y_c \cdot A}$$

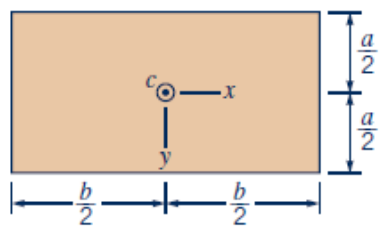
$$= y_c + \frac{I_{xc}}{y_c \cdot A}$$

similarly,
$$x_R = \frac{\int (x \cdot y) dA}{y_c \cdot A} = x_c + \frac{I_{xyc}}{y_c \cdot A}$$



2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- 기하학적 특성 Geometric properties



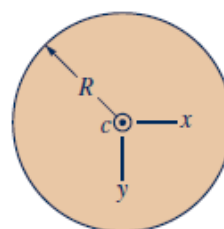
(a) Rectangle

$$A = ba$$

$$I_{xc} = \frac{1}{12} ba^3$$

$$I_{yc} = \frac{1}{12} ab^3$$

$$I_{xyc} = 0$$

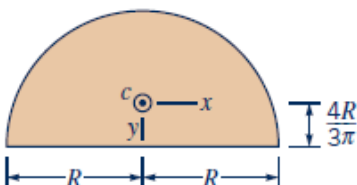


(b) Circle

$$A = \pi R^2$$

$$I_{xc} = I_{yc} = \frac{\pi R^4}{4}$$

$$I_{xyc} = 0$$



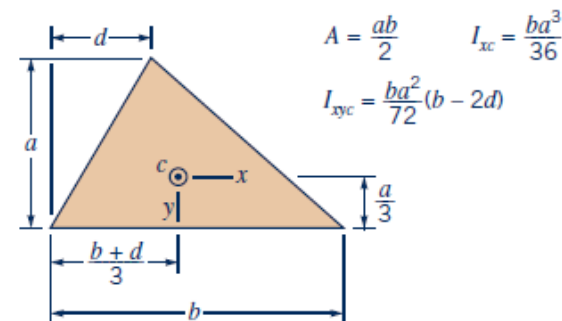
(c) Semicircle

$$A = \frac{\pi R^2}{2}$$

$$I_{xc} = 0.1098R^4$$

$$I_{yc} = 0.3927R^4$$

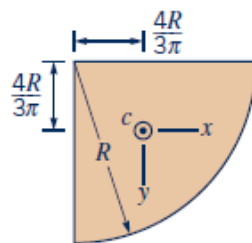
$$I_{xyc} = 0$$



(d) Triangle

$$A = \frac{ab}{2} \quad I_{xc} = \frac{ba^3}{36}$$

$$I_{xyc} = \frac{ba^2}{72}(b - 2d)$$



(e) Quarter circle

$$A = \frac{\pi R^2}{4}$$

$$I_{xc} = I_{yc} = 0.05488R^4$$

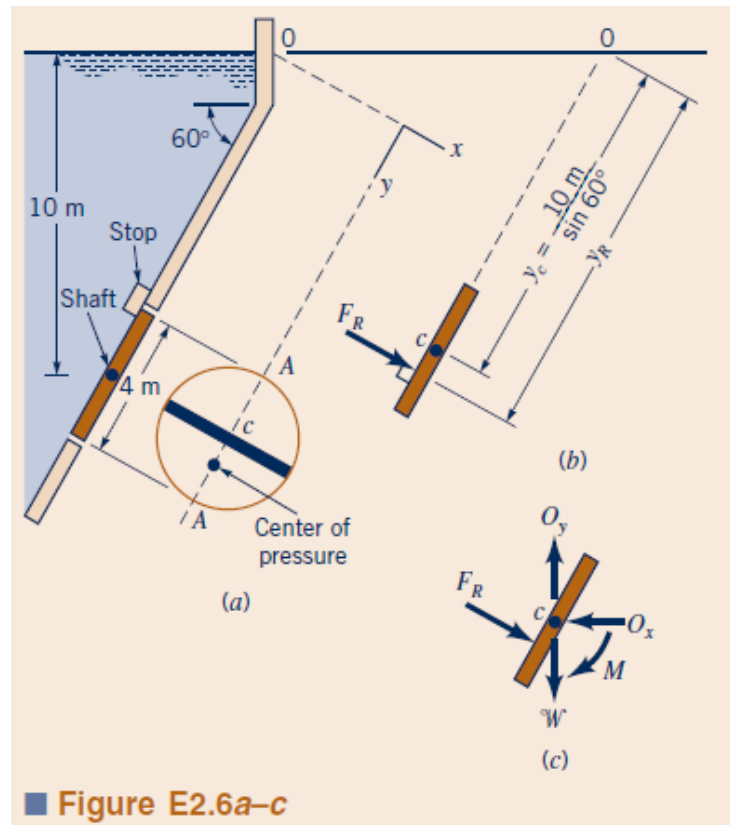
$$I_{xyc} = -0.01647R^4$$

■ Figure 2.18 Geometric properties of some common shapes.



2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- Ex 2.6) Determine (a) the magnitude and location of the resultant force exerted on the gate by the water and (b) the moment that would have to be applied to the shaft to open the gate. (a) 물이 수문에 가하는 합력의 크기와 (b) 수문을 열기 위해 필요한 모멘트를 구하라





2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- Ex 2.6) Determine (a) the magnitude and location of the resultant force exerted on the gate by the water and (b) the moment that would have to be applied to the shaft to open the gate. (a) 물이 수문에 가하는 합력의 크기와 (b) 수문을 열기 위해 필요한 모멘트를 구하라

(a) 정수력의 크기 Magnitude of the resultant force

$$F_R = p_c \cdot A = \gamma \cdot h_c \cdot A$$

$$= 1000 \left(\frac{kg}{m^3} \right) \times 9.8 \left(\frac{m}{s^2} \right) \times 10(m) \times \pi 2^2 (m^2) = 1.23(MN)$$

정수력의 위치 Location of the resultant force

$x_p = 0$: symmetric 대칭

$$y_R = y_c + \frac{I_{xc}}{y_c \cdot A}$$

$$= \frac{10}{\sin 60} (m) + \frac{\pi R^4 / 4 (m^4)}{\frac{10}{\sin 60} (m) \cdot \pi R^2 (m^2)}$$

$$= 11.68(m)$$

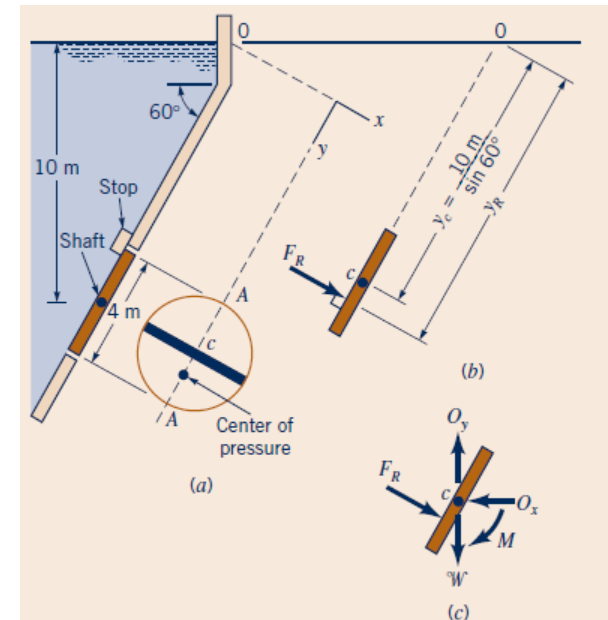


Figure E2.6a-c



2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

(b) 수문을 열기 위해 필요한 모멘트 Moment required to open the gate

$$\sum M_c = 0 \Rightarrow \therefore -M + F_R(y_R - y_c) = 0$$

$$\therefore M = F_R(y_R - y_c) = 1.23(MN) \times \left(11.68 - \frac{10}{\sin 60^\circ} \right) (m)$$

$$= 1.41 \times 10^5 (N \cdot m)$$

hinge

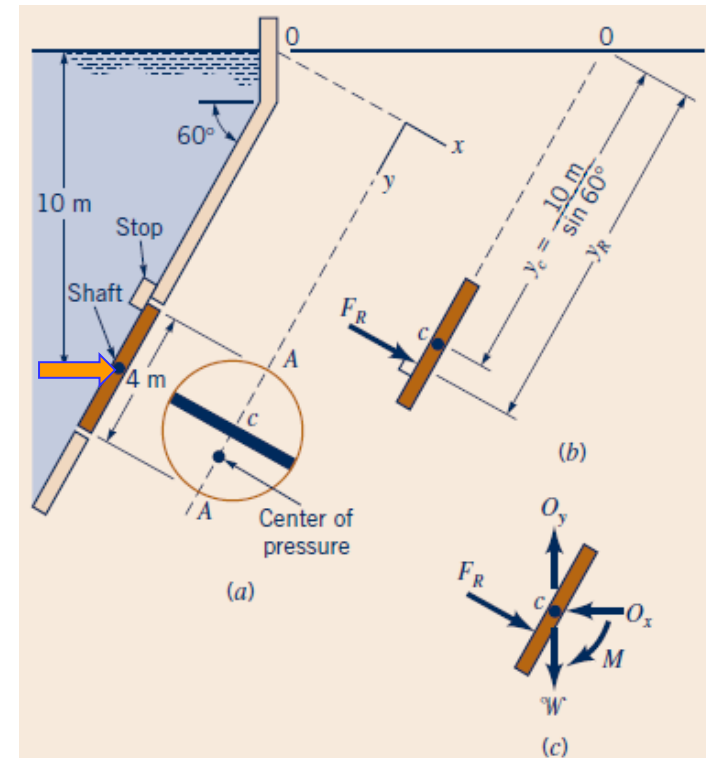
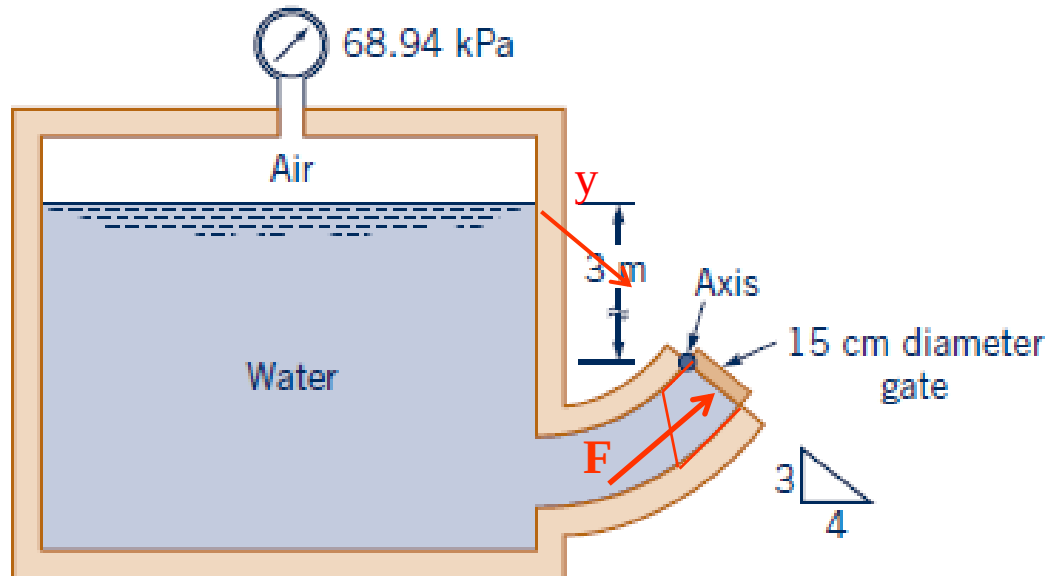


Figure E2.6a-c

Quiz #4: Hydrostatic Force on a Plane Surface

- Q4) The closed vessel of the figure contains water with an air pressure of 68.94(kPa) at the water surface. One side of the vessel contains a spout that is closed by a 15.0 cm diameter circular gate that is hinged along one side as shown. The horizontal axis of the hinge is located 3.0 m below the water surface. Determine the minimum torque that must be applied at the hinge to hold the gate shut. Neglect the weight of the gate and friction at the hinge. 수문을 지탱하기 위해 경첩에 가해야 하는 최소한의 모멘트를 구하라. 단 수문의 무게와 경첩에서의 마찰은 무시하라. (20점)



■ Figure P2.107



2.8 평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

Fi 댐의 수문 개폐 구조?, 운문 댐 구조는? 수문 개폐에 필요한 모터 동력은.....

GPT-5.6 Luna ▾ 유체역학 ...

2.9 압력프리즘 Pressure Prism

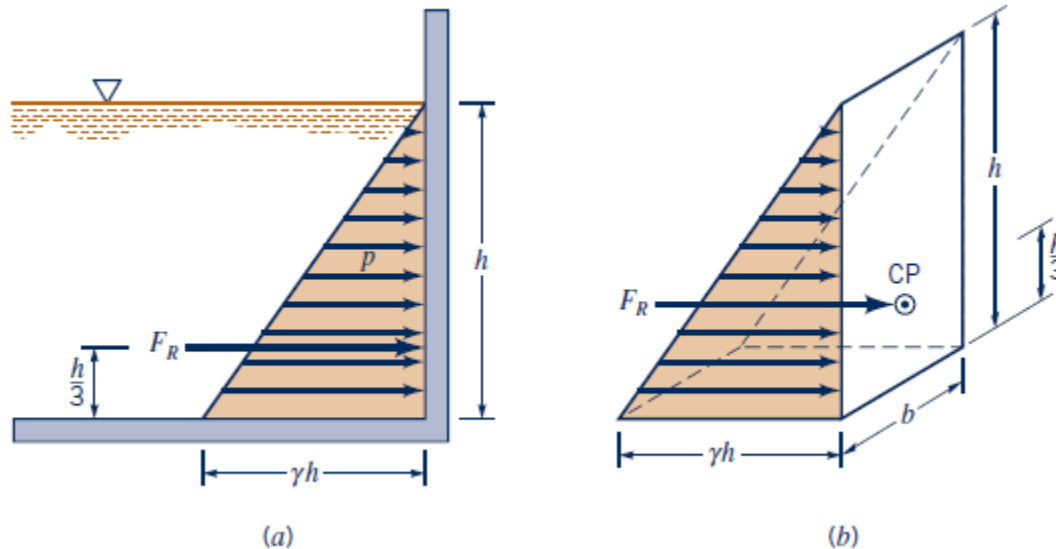
- The magnitude of the resultant fluid force is equal to the volume of the pressure prism and passes through its centroid. 유체의 합력의 크기는 압력프리즘의 체적과 같고, 무게중심을 통과한다

- 크기 **Magnitude**

$$F_R = volume = \frac{(\gamma \cdot h)h}{2} b = \gamma \frac{h}{2} (h \cdot b) = \frac{\gamma \cdot h}{2} A: Fig. 2.19(b)$$

$$= p_c A = (\gamma \cdot h_c) A = \frac{\gamma \cdot h}{2} A: Fig. 2.19(a)$$

- 압력중심 **Center of pressure** = 체적의 도심(무게중심) **centroid of the volume**



■ **Figure 2.19** Pressure prism for vertical rectangular area.

2.9 압력프리즘 Pressure Prism

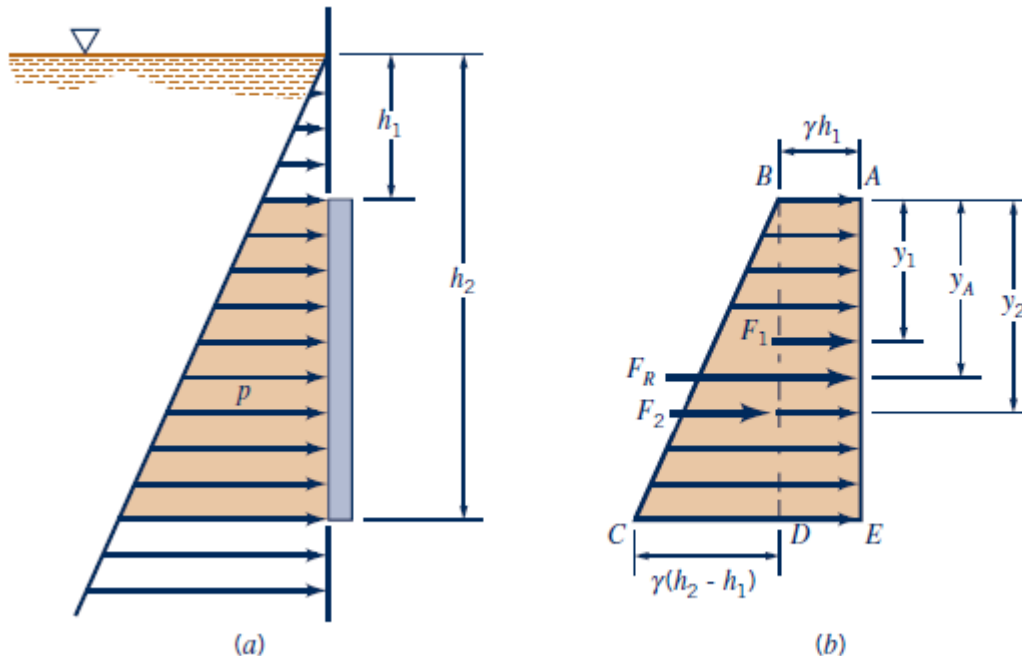
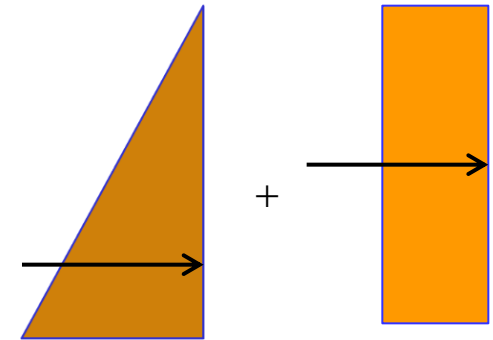
- 크기 **Magnitude**

$$F_R = F_1 + F_2 = \gamma h_1(h_2 - h_1)b + \frac{\gamma(h_2 - h_1)}{2}(h_2 - h_1)b$$

- 압력중심 **Center of pressure** = centroid of the volume

$$F_R y_A = F_1 y_1 + F_2 y_2 = F_1 \frac{h_2 - h_1}{2} + F_2 \frac{2(h_2 - h_1)}{3}$$

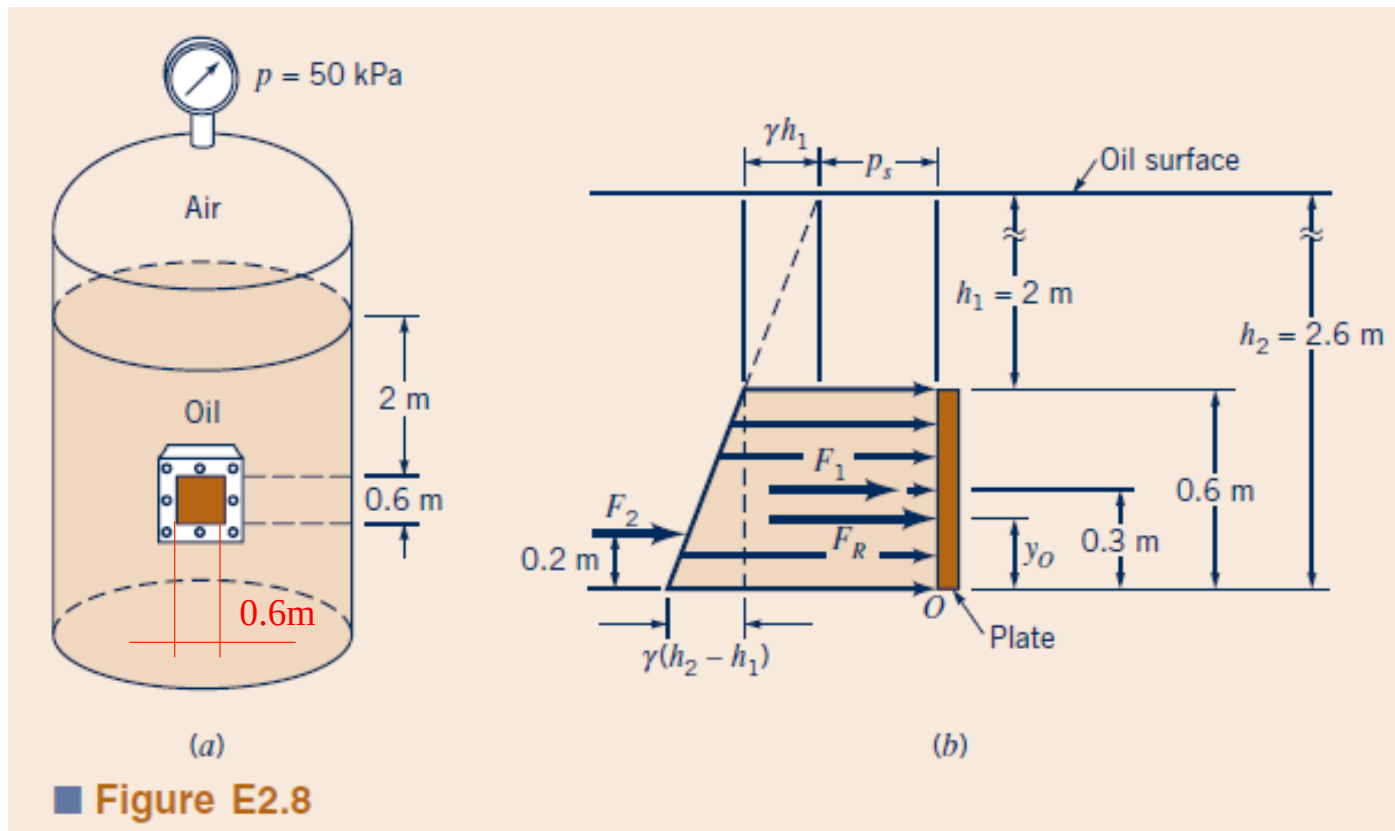
$$\therefore y_A = \frac{F_1}{F_R} \frac{h_2 - h_1}{2} + \frac{F_2}{F_R} \frac{2(h_2 - h_1)}{3}$$



■ **Figure 2.20** Graphical representation of hydrostatic forces on a vertical rectangular surface.

2.9 압력프리즘 Pressure Prism

- Ex 2.8) A pressurized tank contains oil (SG=0.9) and has a square, 0.6m by 0.6m plate to its side. What is the magnitude and location of the resultant force on the attached plate? 오일(SG=0.9)을 채운 가압 탱크에 0.6mX0.6m 크기의 창이 부착되어 있다. 부착한 판에 작용하는 합력의 크기와 위치는?



2.9 압력프리즘 Pressure Prism

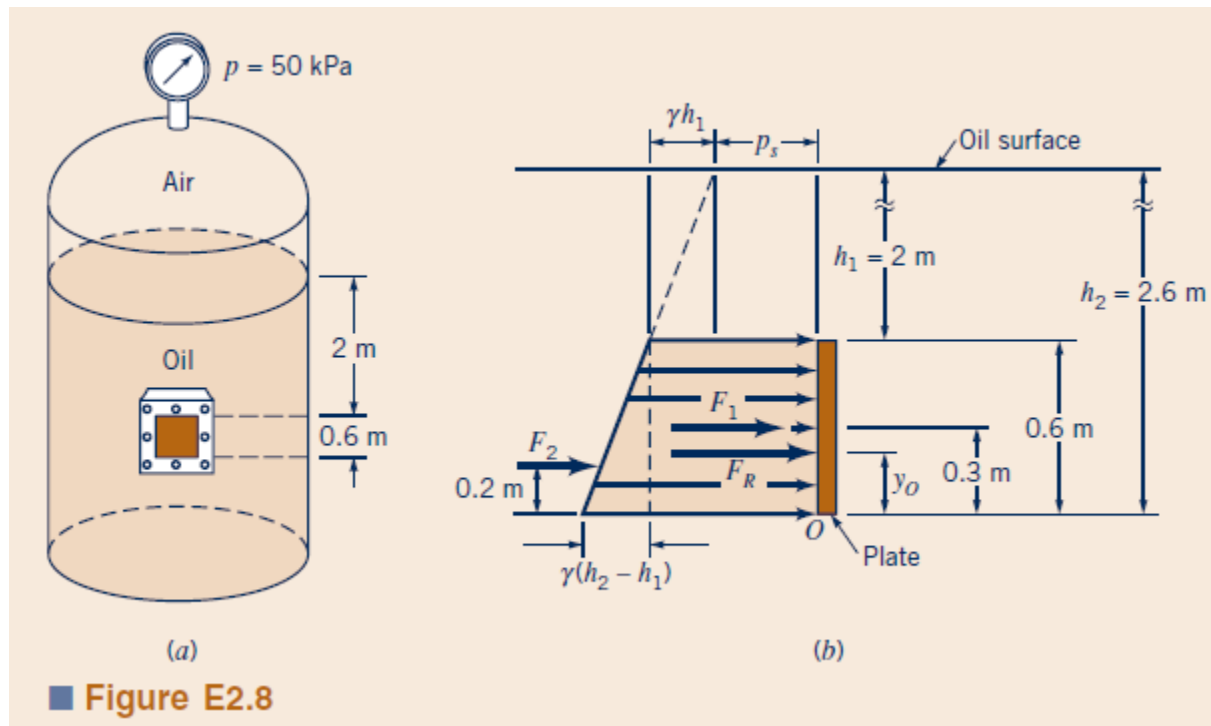
- 크기 Magnitude

$$F_R = F_1 + F_2$$

$$F_1 = [p_s + \gamma h_1]A = [50 \times 10^3 + 0.9 \times 10^3 \times 9.8 \times 2](\text{Pa}) \times 0.6(\text{m}) \times 0.6(\text{m}) = 24.35(\text{kN})$$

$$F_2 = \frac{\gamma(h_2 - h_1)}{2} A = \frac{0.9 \times 10^3 \times 9.8 \times 0.6}{2} (\text{Pa}) \times 0.6(\text{m}) \times 0.6(\text{m}) = 0.953(\text{kN})$$

$$\therefore F_R = F_1 + F_2 = 24.35 + 0.953 = 25.3(\text{kN})$$

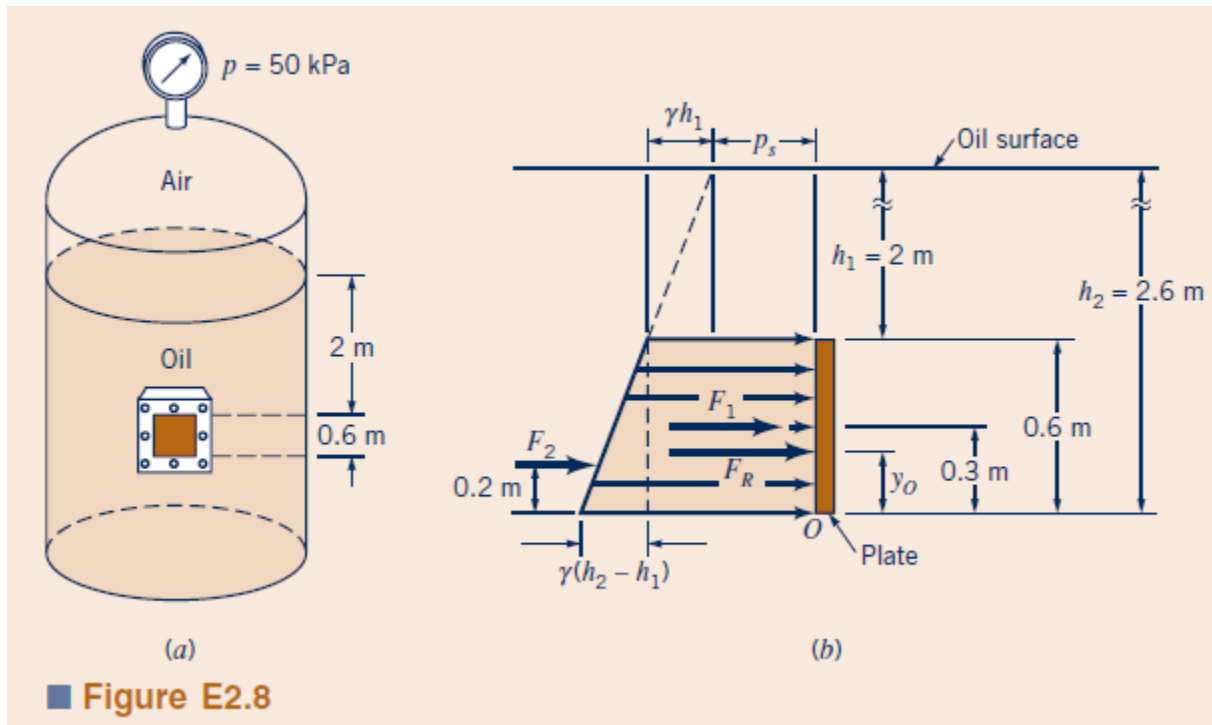


2.9 압력프리즘 Pressure Prism

- 위치 **Location** : o점에 대한 모멘트 **moment about the point o**

$$y_0 F_R = 0.3(m) \times F_1 + 0.2(m) \times F_2 = 0.3 \times 24.35 + 0.2 \times 0.953 \text{ (kN} \cdot \text{m)}$$

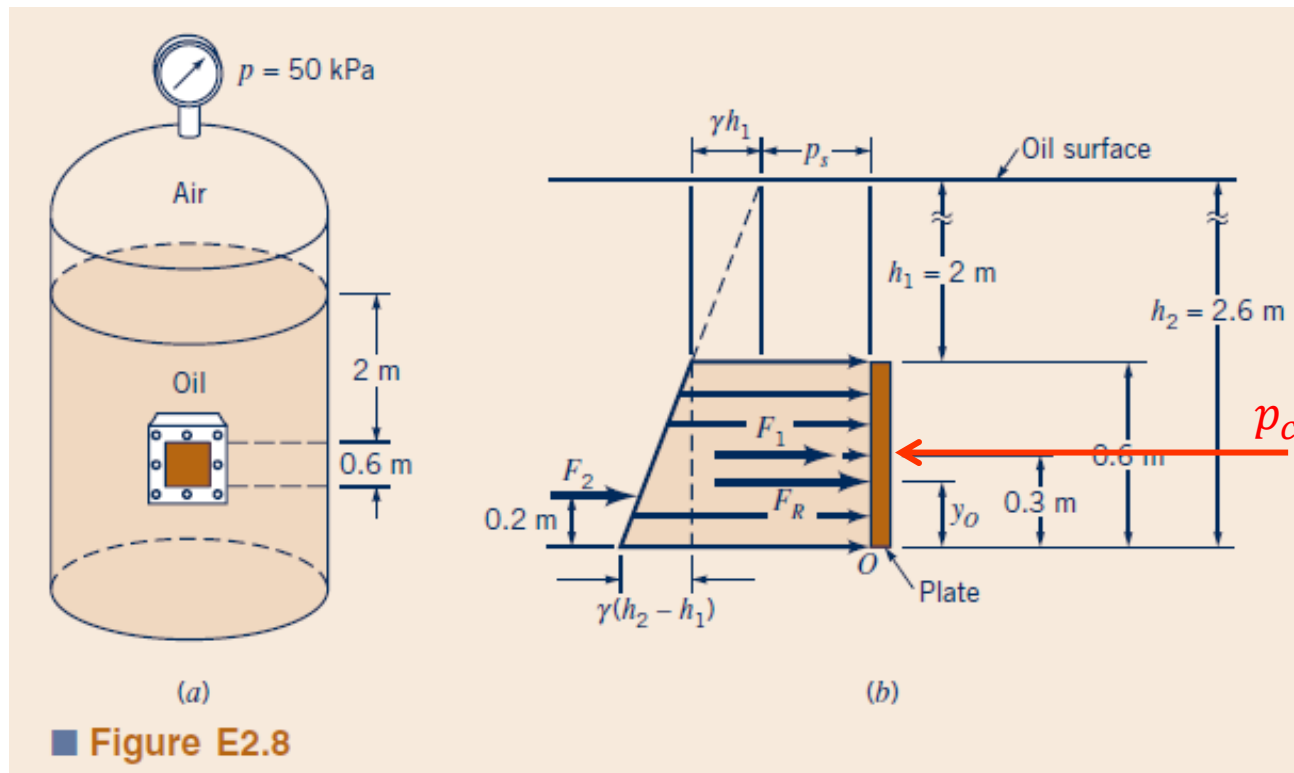
$$\therefore y_0 = \frac{0.3 \times 24.35 + 0.2 \times 0.953}{25.35 \text{ (kN)}} = 0.296 \text{ (m)}$$



평면에 작용하는 정수력 Hydrostatic Force on a Plane Surface

- 크기 Magnitude

$$F_R = p_c A = [50 \times 10^3 + 0.9 \times 10^3 \times 9.8 \times 2.3] (\text{Pa}) \times 0.6 (\text{m}) \times 0.6 (\text{m}) = 25.3 (\text{kN})$$



2.9 압력프리즘 Pressure Prism

- 위치 Location : o점에 대한 모멘트 moment about the point o

$$\begin{aligned} y_R &= y_c + \frac{I_{xc}}{y_c \cdot A} = (2.3)(m) + \frac{ba^3/12 (m^4)}{2.3(m) \cdot ab(m^2)} \\ &= 2.3(m) + \frac{0.6^2/12(m^2)}{2.3(m)} = 2.313(m) \Rightarrow y_0 = 2.6 - y_c = 0.287(m) < 0.296(m) \end{aligned}$$

2.9 압력프리즘 Pressure Prism

- 위치 Location : o점에 대한 모멘트 moment about the point o

$$\begin{aligned} F \cdot y_R &= \int y dF = \int y p dA = \int y (p_{tank} + \gamma y \sin \theta) dA = p_{tank} \int y dA + \int y (\gamma y \sin \theta) dA \\ &= p_{tank} y_c A + (\gamma \sin \theta) I_x = p_{tank} y_c A + (\gamma \sin \theta) (A y_c^2 + I_{xc}) \end{aligned}$$

$$\therefore y_R = \frac{p_{tank} y_c A + (\gamma \sin \theta) (A y_c^2 + I_{xc})}{F}$$

$$= \frac{50 \times (2.3) \times (0.6 \times 0.6) + \frac{0.9 \times 9800}{1000} [(0.6 \times 0.6) (2.3)^2 + \frac{(0.6)(0.6)^3}{12}]}{25.3} = 2.304$$

$$\therefore y_0 = 2.6 - y_R = 2.6 - 2.304 = 0.296(m)$$

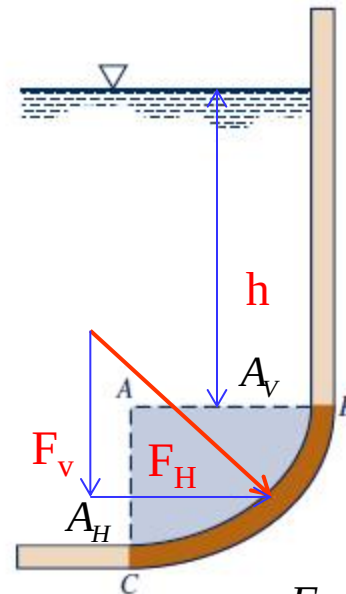
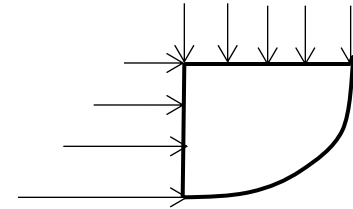
2.10 곡면에 작용하는 정수력 Hydrostatic Force on a Curved Surface

- 수평력과 수직력 Horizontal and vertical forces

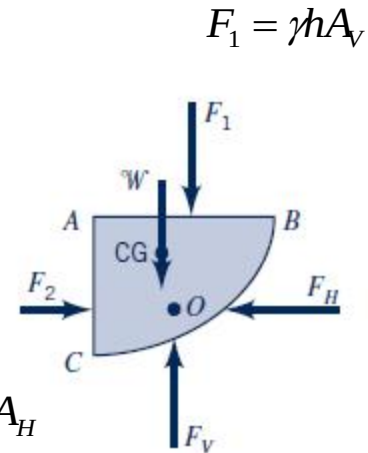
$$F_H = F_2 \text{ and } F_V = F_1 + W$$

- 합력 Resultant force

$$F_R = \sqrt{F_1^2 + F_2^2}$$



$$F_2 = p_C A_H$$



$$F_R = \sqrt{(F_H)^2 + (F_V)^2}$$

(d)



(a)

(b)

(c)

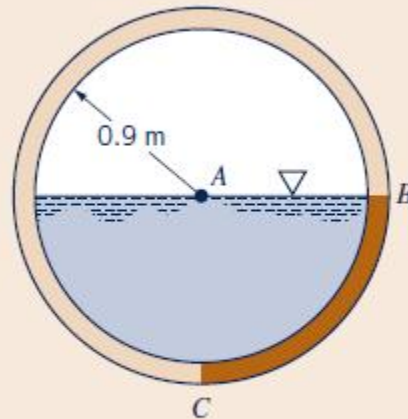
■ **Figure 2.23** Hydrostatic force on a curved surface. (Photograph courtesy of Intex Marketing, Ltd.)

2.10 곡면에 작용하는 정수력 Hydrostatic Force on a Curved Surface

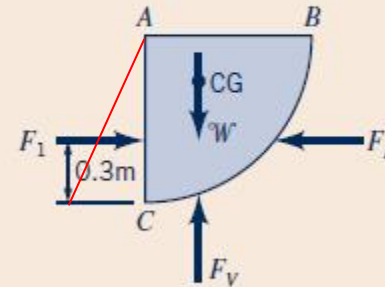
- Ex 2.9) Determine the magnitude and line of action of the resultant force that the water exerts on a 0.3(m) length of the curved section BC of the conduit wall. 길이 0.3m인 곡관의 BC 곡면에 작용하는 물에 의한 힘의 크기와 합력의 작용선(line of action)을 구하라



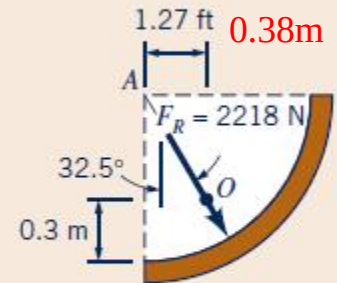
(a)



(b)



(c)



(d)

■ **Figure E2.9** (Photograph courtesy of CONTECH Construction Products, Inc.)

2.10 곡면에 작용하는 정수력 Hydrostatic Force on a Curved Surface

- 수평력 Horizontal force

$$F_H = F_1 = \frac{\gamma R}{2} RL = \frac{1000 \left(\frac{kg}{m^3} \right) \times 9.8 \left(\frac{m}{s^2} \right) \times 0.9(m)}{2} \times 0.9(m) \times 0.3(m) = 1191(N)$$

- 수평력의 압력중심 Center of pressure of the horizontal force

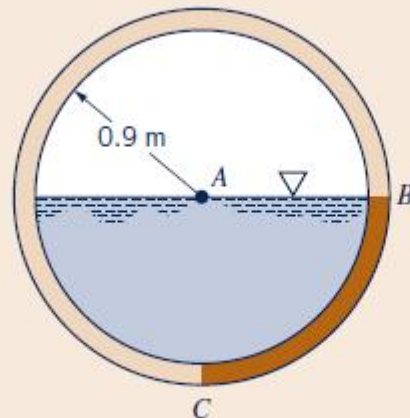
$$y_p = \frac{R}{3} = \frac{0.9}{3} = 0.3(m): \frac{R}{2} + \frac{L(R)^{3/2}}{(R/2)RL} = \frac{R}{2} + \frac{R}{6} = \frac{2R}{3} \text{ from surface}$$

- 수직력 Vertical force

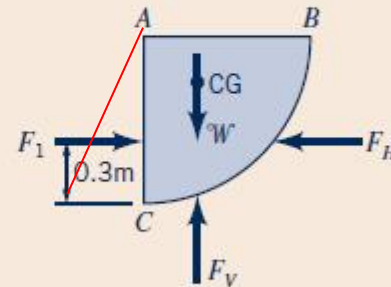
$$F_V = W = \gamma V = \frac{\gamma \pi R^2}{4} L = \frac{1000 \left(\frac{kg}{m^3} \right) \times 9.8 \left(\frac{m}{s^2} \right) \pi \times [0.9(m)]^2}{4} \times 0.3(m) = 1870(N)$$



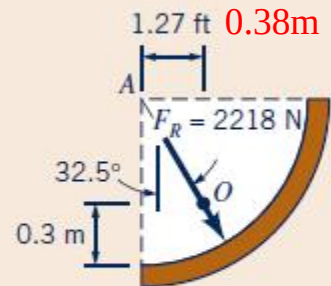
(a)



(b)



(c)



(d)

2.10 곡면에 작용하는 정수력 Hydrostatic Force on a Curved Surface

- 합력의 크기 Magnitude of the resultant force

$$F_R = \sqrt{F_1^2 + F_2^2} = \sqrt{1191^2 + 1870^2} (N) = 2281 (N)$$

- 무게중심 Centroid

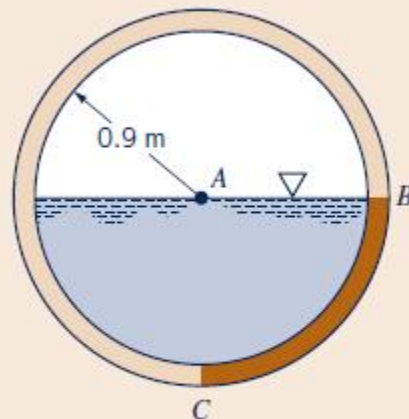
$$x_c = \frac{4R}{3\pi} = \frac{4 \times 0.9 (m)}{3\pi} = 0.38 (m)$$

- 합력의 작용선 Line of action of the resultant force

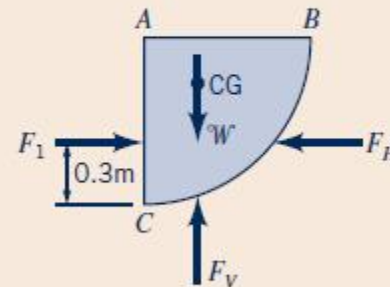
$$\theta = \tan^{-1} \frac{F_H}{F_V} = \tan^{-1} \frac{1191}{1870} = 32.5(^{\circ})$$



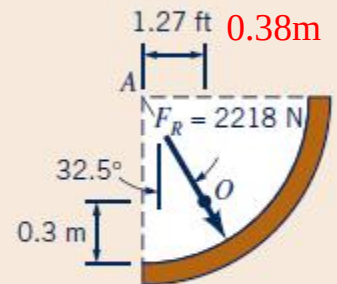
(a)



(b)



(c)

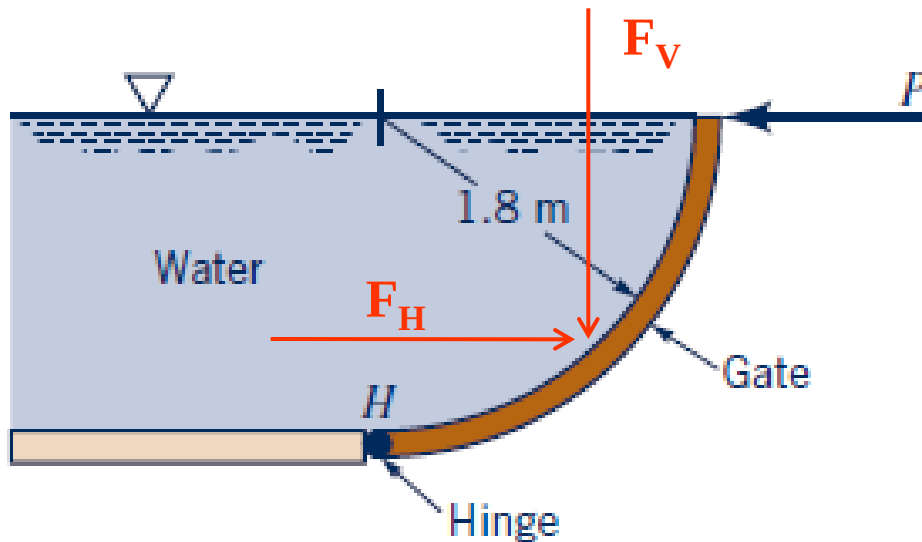


(d)

■ Figure E2.9 (Photograph courtesy of CONTECH Construction Products, Inc.)

Quiz #5 : review problem

- Q5) The 5.5(m) long gate of the figure is a quarter circle and is hinged at H. Determine the horizontal force P , required to hold the gate in place. Neglect friction at the hinge and the weight of the gate. 아래 그림은 $\frac{1}{4}$ 원형 수문으로 길이가 5.5m이며 H점에 경첩으로 연결되어 있다. 수문을 지탱하기 위한 수평 힘 P 를 구하라. 경첩의 마찰과 수문의 무게는 무시하라. (30점)

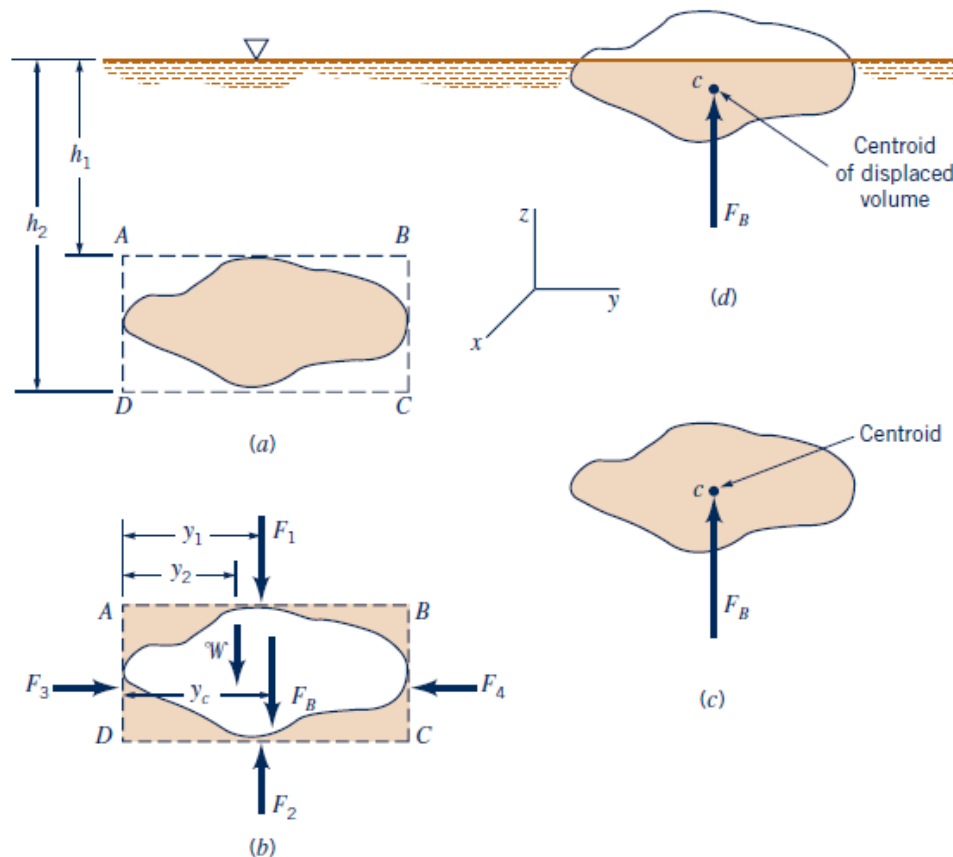


■ Figure P2.117

2.11 부력, 부유 그리고 안정성 Buoyancy, Flotation and Stability

• 2.11.1 아르키메데스 원리 Archimedes' Principle

부력의 크기는 물체에 의해 배제된 유체의 무게와 같고 수직방향을 향한다. The buoyant force has a magnitude equal to the weight of the fluid displaced by the body and is directed vertically upward.



■ Figure 2.24 Buoyant force on submerged and floating bodies.

2.11 부력, 부유 그리고 안정성 Buoyancy, Flotation and Stability

2.11.1 아르키메데스 원리 Archmedes' Principle

magnitude 크기

$$F_B = F_2 - F_1 - W$$

$$F_2 - F_1 = \gamma(h_2 - h_1)A$$

$$W = \gamma[(h_2 - h_1)A - V(\text{volume of body})]$$

$$\therefore F_B = \gamma V = \gamma \cdot \text{volume of body}$$

location = centroid 위치=무게중심

$$F_B y = (F_2 - F_1)y_1 - W \cdot y_2$$

$$\therefore (\gamma \cdot V)y = \gamma(h_2 - h_1)Ay_1 - \gamma[(h_2 - h_1)A - V]y_2$$

$$\therefore y = \frac{(h_2 - h_1)Ay_1 - [(h_2 - h_1)A - V]y_2}{V}$$

$$= \frac{V_{total}y_1 - [V_{total} - V]y_2}{V} = \text{centroid of the body} = y_c$$

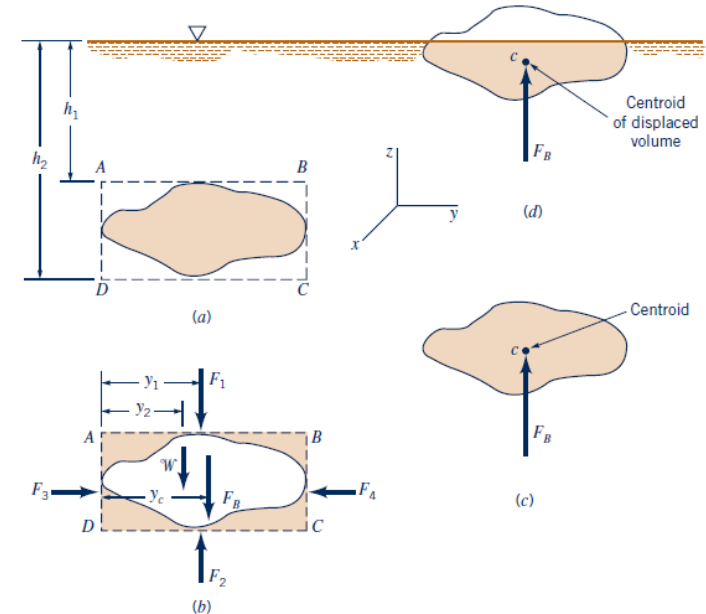


Figure 2.24 Buoyant force on submerged and floating bodies.

2.11 부력, 부유 그리고 안정성 Buoyancy, Flotation and Stability

Ex2.10) A Type I offshore life jacket (personal floating device) of the type worn by commercial fishermen is shown in the Fig. E2.10a. It is designed for extended survival in rough, open water. According to U.S. Coast Guard regulations, the life jacket must provide a minimum 98(N) net upward force on the user. Consider such a life jacket that uses a foam material with a specific weight of $314.3(\text{N/m}^3)$ for the main floatation material. The remaining material (cloth, straps, fasteners, etc.) weighs 5.8(N) and is of negligible volume. Determine the minimum volume of foam for this life jacket. 이 구명 자켓의 최소 체적을 구하라.



Figure E2.10a

2.11 부력, 부유 그리고 안정성 Buoyancy, Flotation and Stability

Ex2.10) 이 구멍 자켓의 최소 체적을 구하라.

magnitude 크기

Lift jacket이 완전히 물에 잠긴다고 가정

$$F_B = W_{foam} + W_{remaining} + W_{user}$$

$$F_B = \gamma_{water} V$$

$$W_{foam} = \gamma_{foam} V$$

$$W_{remaining} = 5.8(N)$$

$$W_{user} = 98(N)$$

$$\therefore \gamma_{water} V = \gamma_{foam} V + 5.8 + 98$$

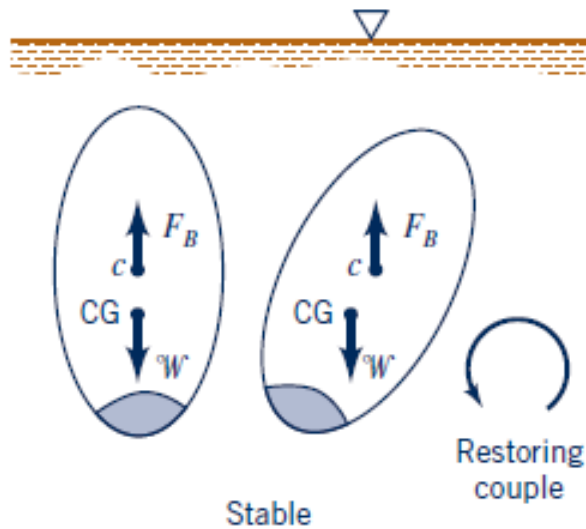
$$\therefore V = \frac{103.8(N)}{\gamma_{water} - \gamma_{foam}} = \frac{103.8(N)}{1000 \left(\frac{kg}{m^3} \right) \times 9.8 \left(\frac{m}{s^2} \right) - 314.3 \left(\frac{N}{m^3} \right)} = 0.011(m^3)$$



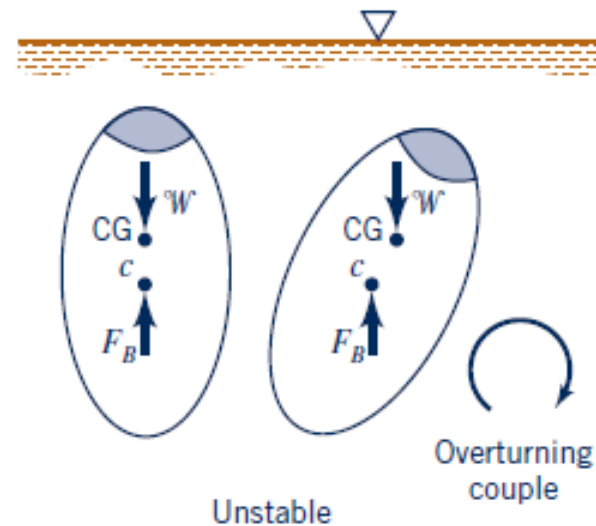
Figure E2.10a

2.11 부력, 부유 그리고 안정성 Buoyancy, Flotation and Stability

- 2.11.2 안정성 Stability



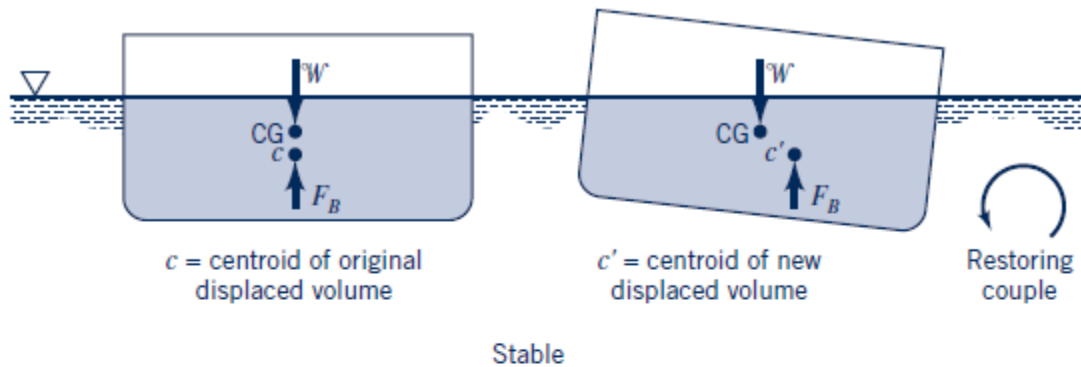
■ **Figure 2.25** Stability of a completely immersed body—center of gravity below centroid.



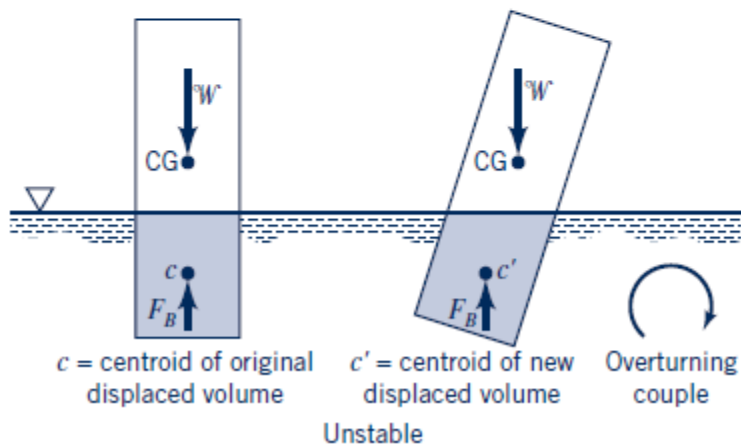
■ **Figure 2.26** Stability of a completely immersed body—center of gravity above centroid.

2.11 부력, 부유 그리고 안정성 Buoyancy, Flotation and Stability

- 2.11.2 Stability



■ **Figure 2.27** Stability of a floating body—stable configuration.



■ **Figure 2.28** Stability of a floating body—unstable configuration.

Ex) buoy



2.12 강체운동을 하는 유체 속 압력 변화

Pressure Variation in a Fluid with Rigid Body Motion

- Pressure variation for fluids in motion 운동하는 유체 속 압력 변화

$$-\nabla p + \rho \vec{g} = \rho \vec{a} \Rightarrow \nabla p = -\rho \vec{a} + \rho \vec{g}$$

- 용기 내 압력 변화

$$\begin{aligned} dp &= \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \\ &= -\rho(a_x - g_x)dx - \rho(a_y - g_y)dy - \rho(a_z - g_z)dz \end{aligned}$$

- If $\vec{g} = -g\vec{k}$

$$-\nabla p - \rho \vec{k} = \rho \vec{a}$$

$$\therefore -\frac{\partial p}{\partial x} = \rho a_x, \quad -\frac{\partial p}{\partial y} = \rho a_y \quad \text{and} \quad -\frac{\partial p}{\partial z} = \rho g + \rho a_z = \gamma + \rho a_z$$

$$\begin{aligned} dp &= \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \\ &= -\rho(a_x)dx - \rho(a_y)dy - \rho(a_z + g)dz \end{aligned}$$

2.12 강체운동을 하는 유체 속 압력 변화

Pressure Variation in a Fluid with Rigid Body Motion

• 2.12.1 선형 운동 Linear Motion

- If an open container of a liquid is translating with a constant acceleration $\vec{a}$

since $a_x = 0$

$$-\frac{\partial p}{\partial y} = \rho a_y \quad \text{and} \quad -\frac{\partial p}{\partial z} = \rho g + \rho a_z$$

유체가 담긴 뚜껑이 열린 용기가
일정 가속도 $\vec{a}$ 로 직선 운동을 할 경우

- pressure variation (용기 내) 압력변화

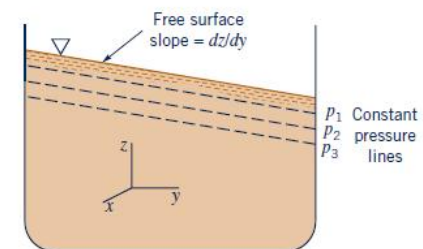
$$\begin{aligned} dp &= \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz = \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz \\ &= -\rho a_y dy - \rho(g + a_z) dz \end{aligned}$$

- Constant pressure lines
(free surface) 등압선

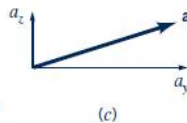
$$\begin{aligned} dp &= 0 \Rightarrow -\rho a_y dy - \rho(g + a_z) dz = 0 \\ \therefore \frac{dz}{dy} &= -\frac{a_y}{g + a_z} \end{aligned}$$



(a)



(b)



(c)

■ Figure 2.29 Linear acceleration of a liquid with a free surface.



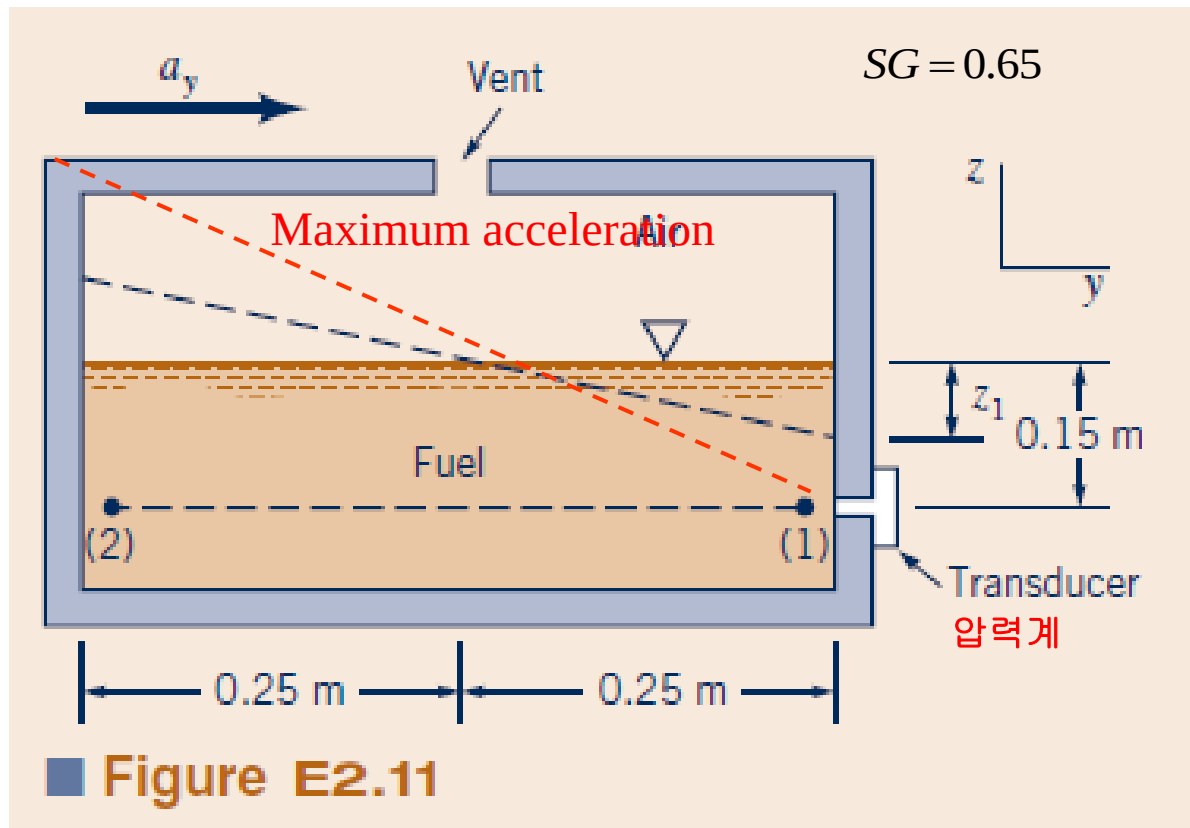
2.12 강체운동을 하는 유체 속 압력 변화

Pressure Variation in a Fluid with Rigid Body Motion

- Ex 2.11) Pressure Variation in an Acceleration Tank

(a) 가속도 a_y 와 압력계 압력의 관계식을 구하라 Determine an expression that relates a_y and the pressure at the transducer.

(b) 연료가 변환기 아래에 떨어지는 최대 가속도는? What is the maximum acceleration that can be occurred before the fuel level drops below the transducer?





2.12 강체운동을 하는 유체 속 압력 변화

Pressure Variation in a Fluid with Rigid Body Motion

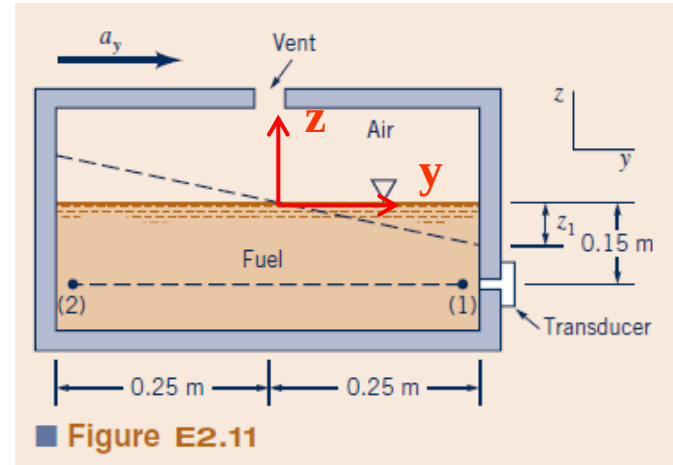
- (a) 가속도 a_y 와 압력계 압력의 관계식을 구하라 Determine an expression that relates a_y and the pressure at the transducer.
- 자유표면 Free surface

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz = -\rho a_y dy - \rho(g + a_z) dz$$

$$\therefore \frac{dz}{dy} = -\frac{a_y}{g + a_z} \Rightarrow z = -\frac{a_y}{g + a_z} y + c \Rightarrow c = 0$$

$$\text{point } (0.25, -z_1) \Rightarrow \therefore z_1 = 0.25 \frac{a_y}{g}$$

- 압력분포 pressure variation



$$\therefore \int_{p_{atm}}^p dp = - \int_{-z_1}^{-0.15} \rho(g + a_z) dz \Rightarrow p - p_{atm} = \rho g(0.15 - z_1)$$

$$\therefore p = p_{atm} + \rho g(0.15 - z_1) \text{ or } p_{gage} = \rho g(0.15 - z_1)$$

$$\begin{aligned} \therefore p_{gage} &= \rho g \left(0.15 - 0.25 \frac{a_y}{g} \right) = 0.65 \times 1000 \left(\frac{kg}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right) \left(0.15 - 0.25 \frac{a_y}{g} \right) \\ &= \left(955 - 1592.5 \frac{a_y}{g} \right) (pa) \text{ for } z \leq 0.15(m) \end{aligned}$$



2.12 강체운동을 하는 유체 속 압력 변화

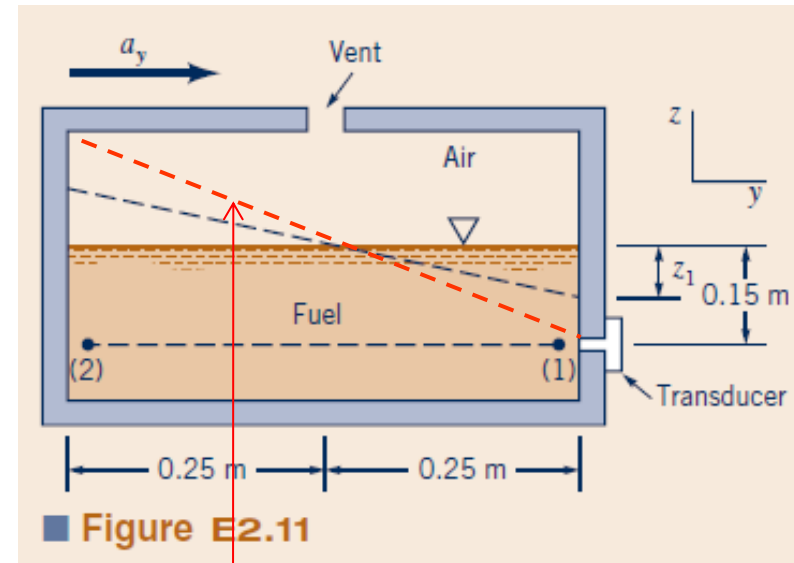
Pressure Variation in a Fluid with Rigid Body Motion

- (b) 연료가 변환기 아래에 떨어지는 최대 가속도는? What is the maximum acceleration that can be occurred before the fuel level drops below the transducer?
- 최대 가속도 maximum acceleration

maximum at $z_1 = 0.15(m)$

$$z_1 = 0.25 \frac{a_y}{g} \Rightarrow 0.15 = 0.25 \frac{a_y}{g}$$

$$\therefore a_y = \frac{0.15}{0.25} 9.8 \left(\frac{m}{s^2} \right) = 5.8 \left(\frac{m}{s^2} \right)$$



Maximum acceleration



2.12 강체운동을 하는 유체 속 압력 변화

Pressure Variation in a Fluid with Rigid Body Motion



자동차 연료 탱크의 수위 측정 방법?

2.13 강체 회전 Rigid Body Rotation

- 일정 회전수로 회전하는 용기 속 유체 내 압력 분포 **Pressure Variation in the Fluid Contained in a Tank Rotating at a Constant Speed**

$$-\nabla p + \rho \vec{g} = \rho \vec{a}$$

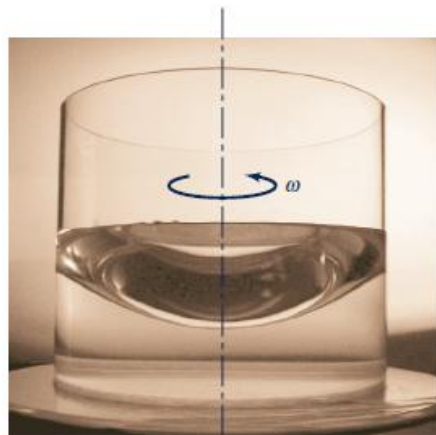
원통 좌표계

$$\nabla p = \frac{\partial p}{\partial r} \vec{e}_r + \frac{\partial p}{r \partial \theta} \vec{e}_\theta + \frac{\partial p}{\partial z} \vec{e}_z: \text{cylindrical coordinate system}$$

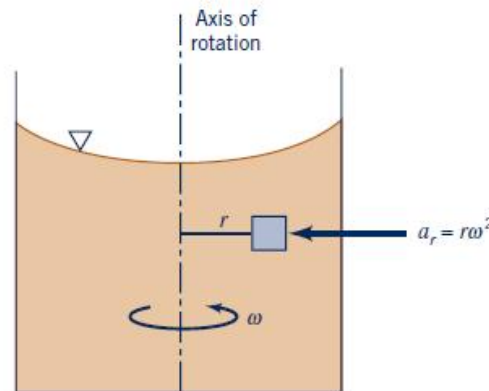
$$\vec{a} = -r\omega^2 \vec{e}_r$$

$$\therefore \frac{\partial p}{\partial r} = \rho r \omega^2, \quad \frac{\partial p}{r \partial \theta} = 0 \quad \text{and} \quad \frac{\partial p}{\partial z} = -\rho g$$

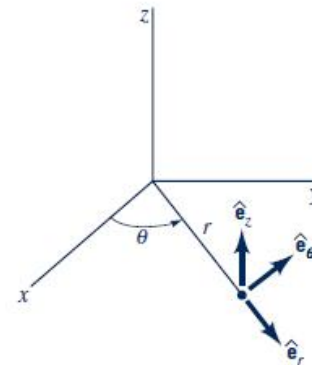
$$dp = \frac{\partial p}{\partial r} dr + \frac{\partial p}{r \partial \theta} (r d\theta) + \frac{\partial p}{\partial z} dz = \rho r \omega^2 dr - \rho g dz$$



(a)



(b)



(c)

■ **Figure 2.30** Rigid-body rotation of a liquid in a tank. (Photograph courtesy of Geno Pawlak.)

2.13 강체 회전 Rigid Body Rotation

- 등압선 Constant Pressure Lines

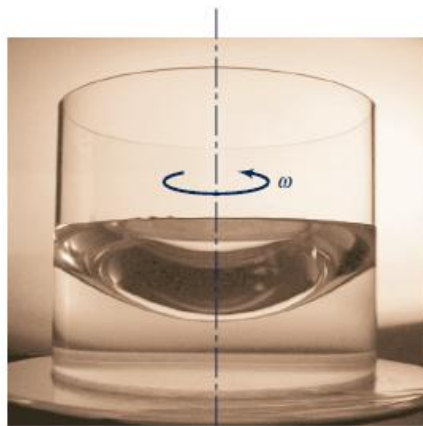
$$dp = 0 \Rightarrow \rho r \omega^2 dr - \rho g dz = 0$$

$$\therefore \frac{dz}{dr} = \frac{r \omega^2}{g} \text{ or } z = \frac{\omega^2 r^2}{2g} + \text{constant}$$

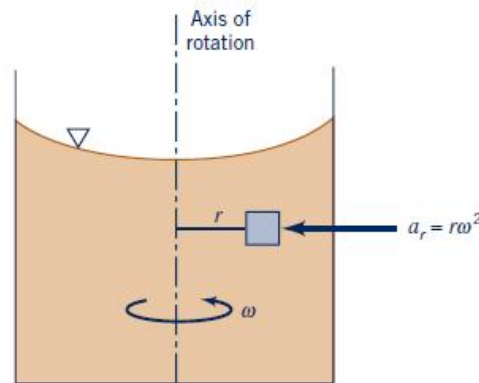
- 압력분포 Pressure Variation

$$\int dp = \int \rho r \omega^2 dr - \int \rho g dz$$

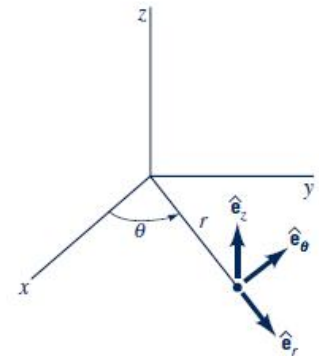
$$\therefore p = \frac{\rho \omega^2 r^2}{2} - \rho g z + \text{constant}$$



(a)



(b)



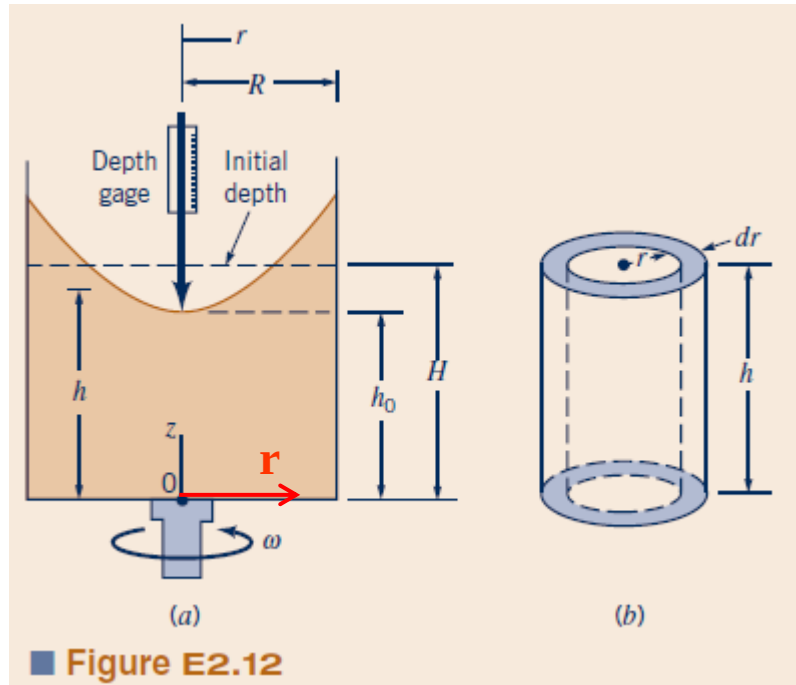
(c)

■ Figure 2.30 Rigid-body rotation of a liquid in a tank. (Photograph courtesy of Geno Pawlak.)

Example : 2.13 강체 회전 Rigid Body Rotation

- Ex 2.12) 회전 용기 속 액체의 자유표면 모양 Free Surface Shape of Liquid in a Rotating Tank

회전 각속도와 유체수위 변화의 관계식을 구하라 Determine the relationship between the depth change in fluid level and the angular velocity.



Example : 2.13 강체 회전 Rigid Body Rotation

- 자유표면 Free Surface

$$dp = 0 \Rightarrow \frac{\partial p}{\partial r} dr + \frac{\partial p}{r \partial \theta} (r d\theta) + \frac{\partial p}{\partial z} dz = \rho r \omega^2 dr - \rho g dz = 0$$

$$\therefore \frac{dz}{dr} = \frac{r \omega^2}{g} \text{ or } z = \frac{\omega^2 r^2}{2g} + \text{constant}$$

$$\therefore h = \frac{\omega^2 r^2}{2g} + h_0$$

- 깊이 변화 Depth change

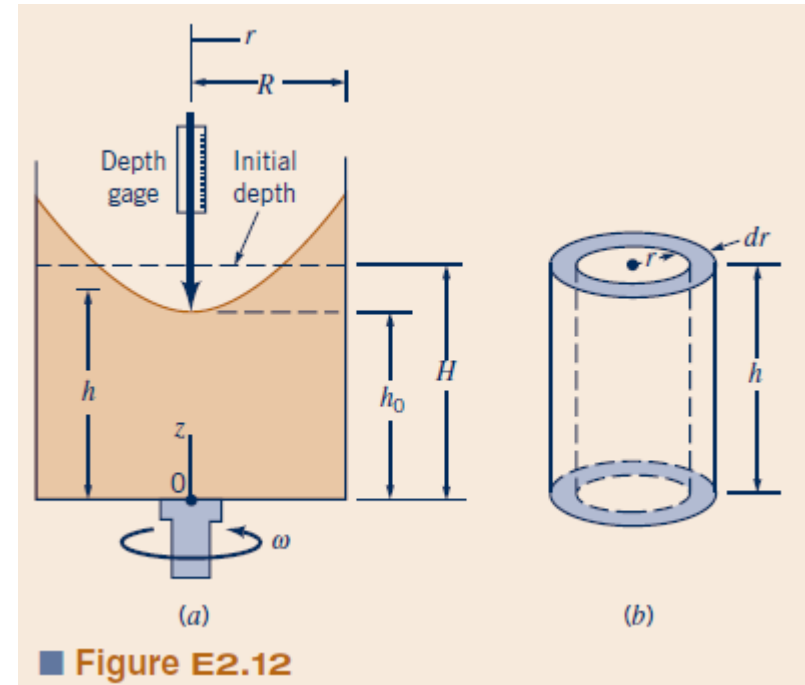
$$V = \text{volume} = \pi R^2 H = \int_0^R (2\pi r) h \cdot dr$$

$$= 2\pi \int_0^R r \left(\frac{\omega^2 r^2}{2g} + h_0 \right) \cdot dr$$

$$= \frac{\pi \omega^2}{g} \int_0^R (r^3) dr + 2\pi h_0 \int_0^R (r) dr$$

$$= \frac{\pi \omega^2 R^4}{4g} + 2\pi h_0 \frac{R^2}{2}$$

$$\therefore H = h_0 + \frac{\omega^2 R^2}{4g}$$



강체 회전 Rigid Body Rotation 산업적 응용



GPT-5.6 Luna ▾

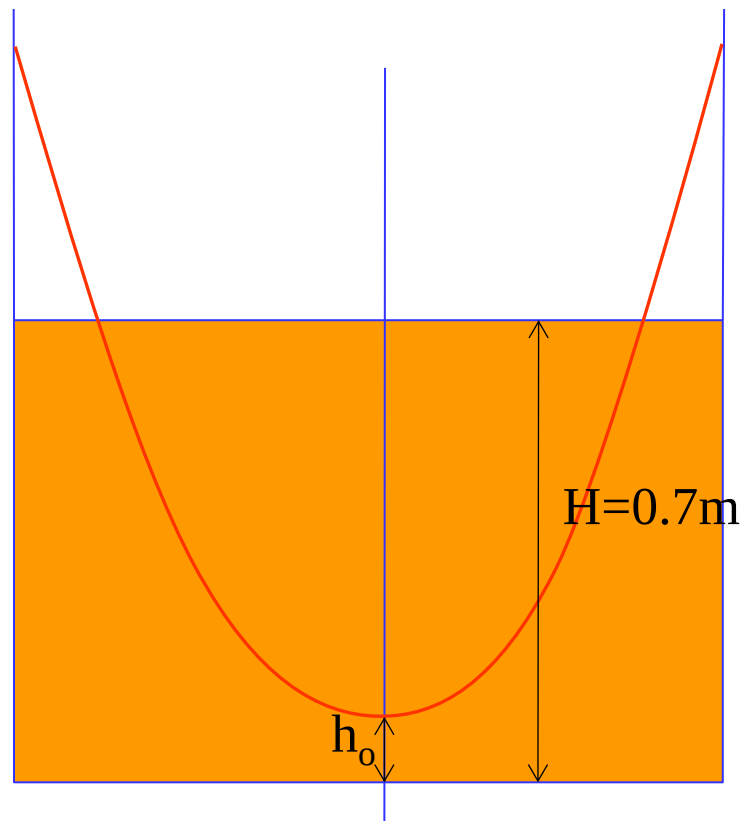
유체역학 ...

강체 회전 산업적 응용 사례?

한 가지를 선택하여 공부하기

2.13 강체 회전 Rigid Body Rotation

- 윗면이 열리고 직경이 1.0(m)인 탱크가 정지 상태에서 높이 0.7(m)의 물을 담고 있다. 탱크가 수직축을 중심으로 회전함에 따라 수면의 중심이 낮아진다. 탱크의 바닥면이 드러나기 시작하는 회전각속도는? 탱크로부터 물은 넘치지 않는다. An open 1(m) diameter tank contains water at a depth of 0.7(m) when at rest. As the tank is rotated about its vertical axis, the center of the fluid surface is depressed. At what angular velocity will the bottom of the tank first be exposed? No water is spilled from the tank. (15점)



강체 회전 Rigid Body Rotation

- 자유표면 Free Surface (5점)

$$h = \frac{\omega^2 r^2}{2g} + h_0: h = h_0 \text{ at } r = 0$$

- 깊이 변화 Depth change (5점)

$$V = \text{volume} = \pi R^2 H = \int_0^R (2\pi r) h \cdot dr$$

$$= 2\pi \int_0^R r \left(\frac{\omega^2 r^2}{2g} + h_0 \right) \cdot dr$$

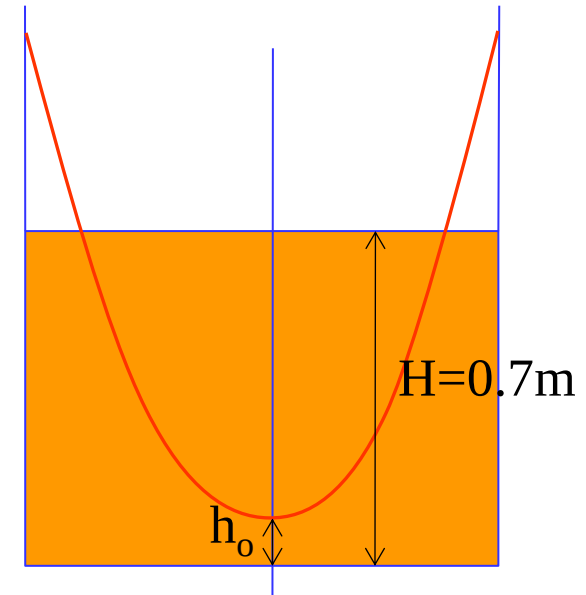
$$= \frac{\pi \omega^2}{g} \int_0^R (r^3) dr + 2\pi h_0 \int_0^R (r) dr$$

$$= \frac{\pi \omega^2 R^4}{4g} + 2\pi h_0 \frac{R^2}{2}$$

$$\therefore H = h_0 + \frac{\omega^2 R^2}{4g}$$

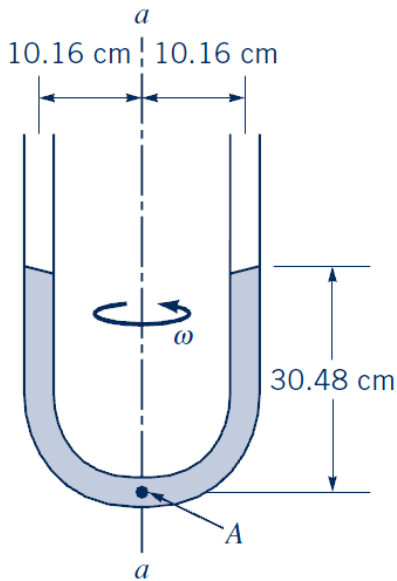
- 회전 각속도 (5점)

$$\omega = \frac{\sqrt{4gH}}{R} = \frac{\sqrt{4 \times 9.8 \left(\frac{m}{s^2} \right) 0.7(m)}}{0.5(m)} = 10.5 \left(\frac{rad}{s} \right)$$



Quiz #6: 2.13 강체 회전 Rigid Body Rotation

- Q6) 물이 일부분 채워진 U-관 이 축 a-a에 대하여 회전한다. 관의 바닥 (A 점)에서 물이 증발하기 시작하는 회전속도를 구하라 The U-tube shown below is partially filled with water and rotates around the axis a-a. Determine the angular velocity that will cause the water to start to vaporize at the bottom of the tube (point A).



- 자유 표면 모양
- 물이 증발하기 위한 조건

■ Figure P2.157

Quiz #6: 2.13 강체 회전 Rigid Body Rotation

- 압력분포 **Pressure variation**

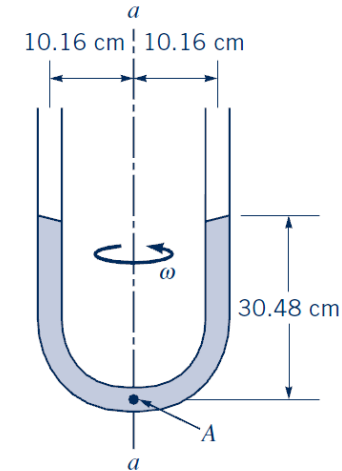
$$dp = \frac{\partial p}{\partial r} dr + \frac{\partial p}{r \partial \theta} (r d\theta) + \frac{\partial p}{\partial z} dz = \rho r \omega^2 dr - \rho g dz$$

$$\int dp = \int \rho r \omega^2 dr - \int \rho g dz$$

$$\therefore p = \frac{\rho \omega^2 r^2}{2} - \rho g z + c$$

$$p = p_{atm} \quad \text{at } r = 0.1016(m) \text{ and } z = 0.3048(m)$$

$$p = p_{vapor} \quad \text{at } r = 0 \text{ and } z = 0$$



■ Figure P2.157

$$p = p_{atm} \quad \text{at } r = 0.1016(m) \text{ and } z = 0.3048(m)$$

$$\therefore 101300(pa) = \frac{1000 \left(\frac{kg}{m^3} \right) \omega^2 0.1016^2}{2} - 1000 \left(\frac{kg}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right) 0.3048(m) + c$$

$$\therefore c = 104278(pa) - \frac{1000 \left(\frac{kg}{m^3} \right) \omega^2 0.1016^2}{2}$$

Quiz #6: 2.13 강체 회전 Rigid Body Rotation

- 각속도 Angular velocity

$$p = p_{\text{vapor}} \text{ at } r = 0 \text{ and } z = 0$$

$$\therefore p_{\text{vapor}} = \frac{1000 \left(\frac{kg}{m^3} \right) \omega^2 0^2}{2} - 1000 \left(\frac{kg}{m^3} \right) 9.8 \left(\frac{m}{s^2} \right) 0(m) + c$$

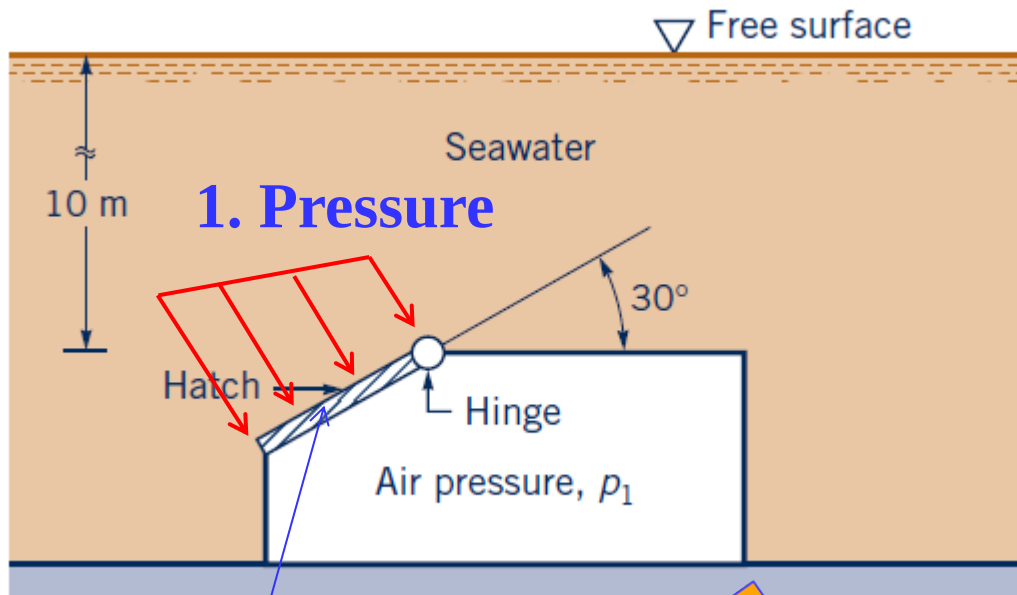
$$p_{\text{vapor}} = 1770(pa)$$

$$\therefore 1770(pa) = c$$

$$= 104278(pa) - \frac{1000 \left(\frac{kg}{m^3} \right) \omega^2 0.1016^2}{2}$$

$$\therefore \omega = \sqrt{\frac{104278 - 1770}{5.16128}} = 140.9 \left(\frac{rad}{s} \right)$$

Check list



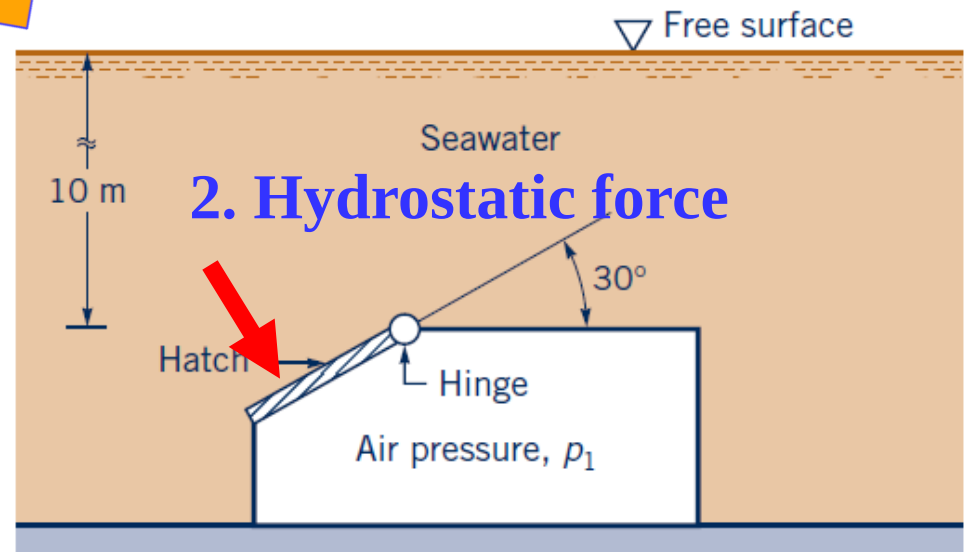
Pressure ?

$$p = p_0 + \gamma h$$

Hydrostatic force

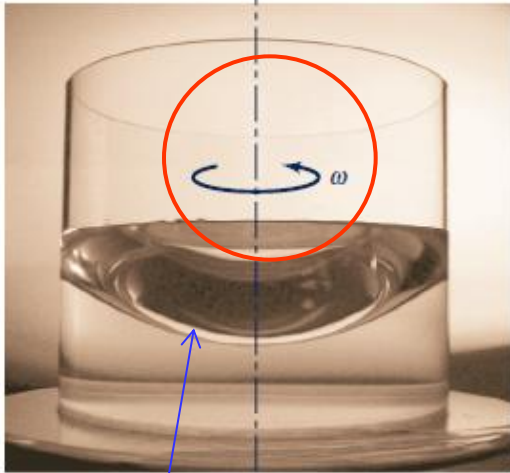
$$F_R = \int dF = p_c \cdot A$$

$$y_R = \frac{\int (y^2) dA}{y_c \cdot A} = y_c + \frac{I_{xc}}{y_c \cdot A}$$

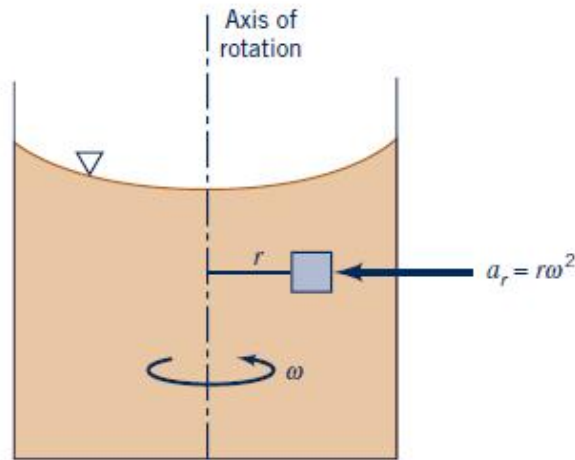


Checklist

Rigid body motion : rotation



(a)



(b)

■ **Figure 2.30** Rigid-body rotation of a liquid in a tank. (Photograph courtesy of Geno Pawlak.)

Free surface ?

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz$$

$$\frac{dz}{dy} = -\frac{a_y}{g + a_z}$$

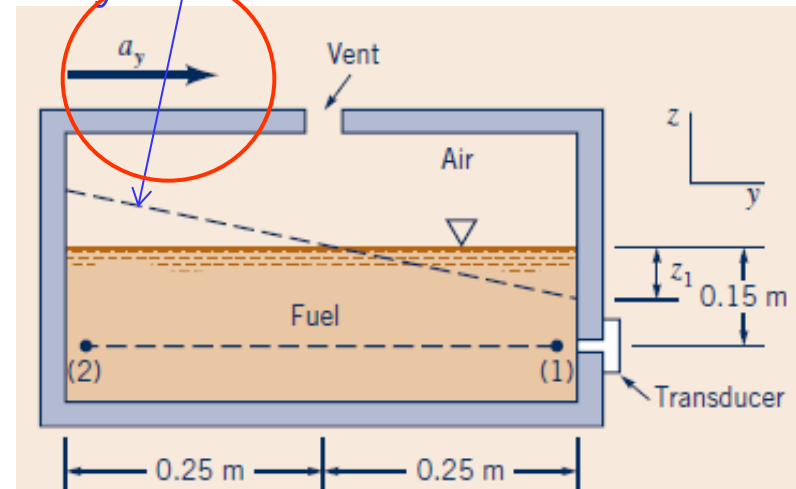
Free surface ?

$$dp = \frac{\partial p}{\partial r} dr + \frac{\partial p}{r \partial \theta} (r d\theta) + \frac{\partial p}{\partial z} dz$$

$$= \rho r \omega^2 dr - \rho g dz$$

$$\therefore \frac{dz}{dr} = \frac{r \omega^2}{g} \text{ or } z = \frac{\omega^2 r^2}{2g} + \text{constant}$$

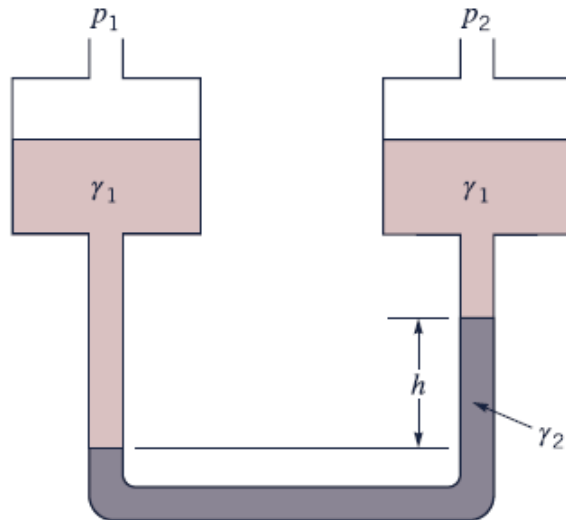
Rigid body motion : linear acceleration



■ **Figure E2.11**

Homework #2

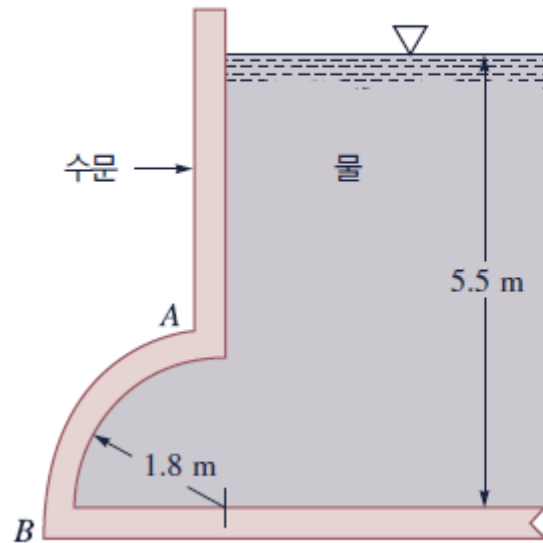
2.15 기체의 작은 압력차를 측정할 경우에는 보통 그림 P2.15와 같은 정밀 액주계(*micromanometer*)가 사용된다. 이 기구는 비중량이 γ_1 인 액체로 채워진 단면적 A_r 인 두 개의 커다란 용기와 이들을 연결하는 단면적 A_t 의 U자관에는 비중량이 γ_2 인 액체가 있다. 기체의 압력차 $p_1 - p_2$ 가 가해지면 눈금차 h 가 발생한다. 작은 압력차에서도 눈금을 쉽게 읽을 수 있도록, 눈금차 h 를 충분히 크게 하는 것이 필요하다. 단면적의 비 A_t/A_r 가 작을 때, h 와 $p_1 - p_2$ 사이의 관계식을 구하라. 또한, 비중량의 차 $\gamma_2 - \gamma_1$ 을 작게 함으로써 눈금차 h 를 확대시킬 수 있음을 보여라. 초기에($p_1 = p_2$ 일 때) 두 용기 안의 액체 높이는 같다고 가정하라.



■ 그림 P2.15

Homework #2

2.45 탱크의 벽면이 그림 P2.45와 같은 모양을 가지고 있다. 1.2 m 길이의 곡면 AB 에 가해지는 물의 수직, 수평 성분의 힘을 구하라.



■ 그림 P2.45